

Mount Kenya University



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REG NO: BSCCS/2024/73317

COURSE;BACHELOR OF SCIENCE IN COMPUTER SCIENCE

UNIT CODE; BIT3208

UNIT NAME;ADVANCED WEB DESIGN AND DEVELOPMENT

LECTURER;MR NYORO MICHEAL

YEAR;3.2

PROJECT;SUPERMARKET INVENTORY

SELECTED TECH;JAVA+MYSQL

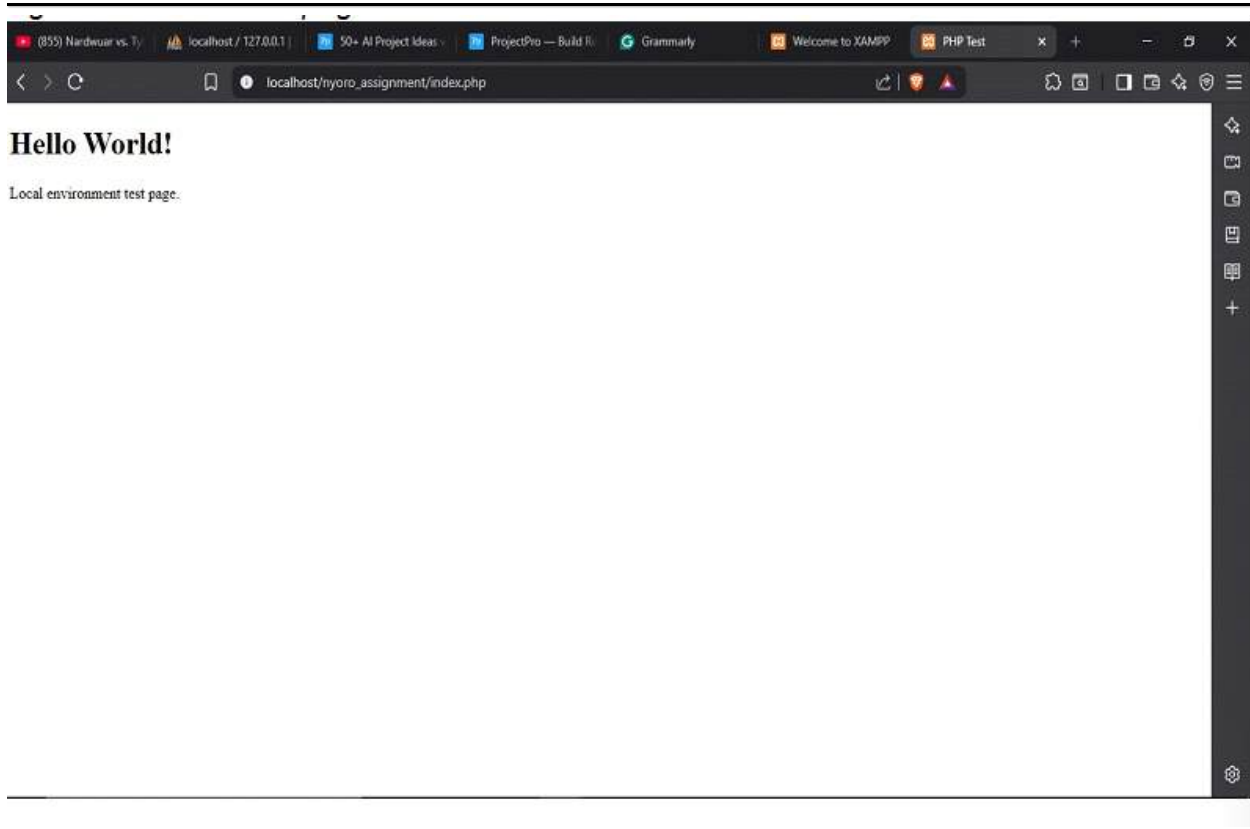
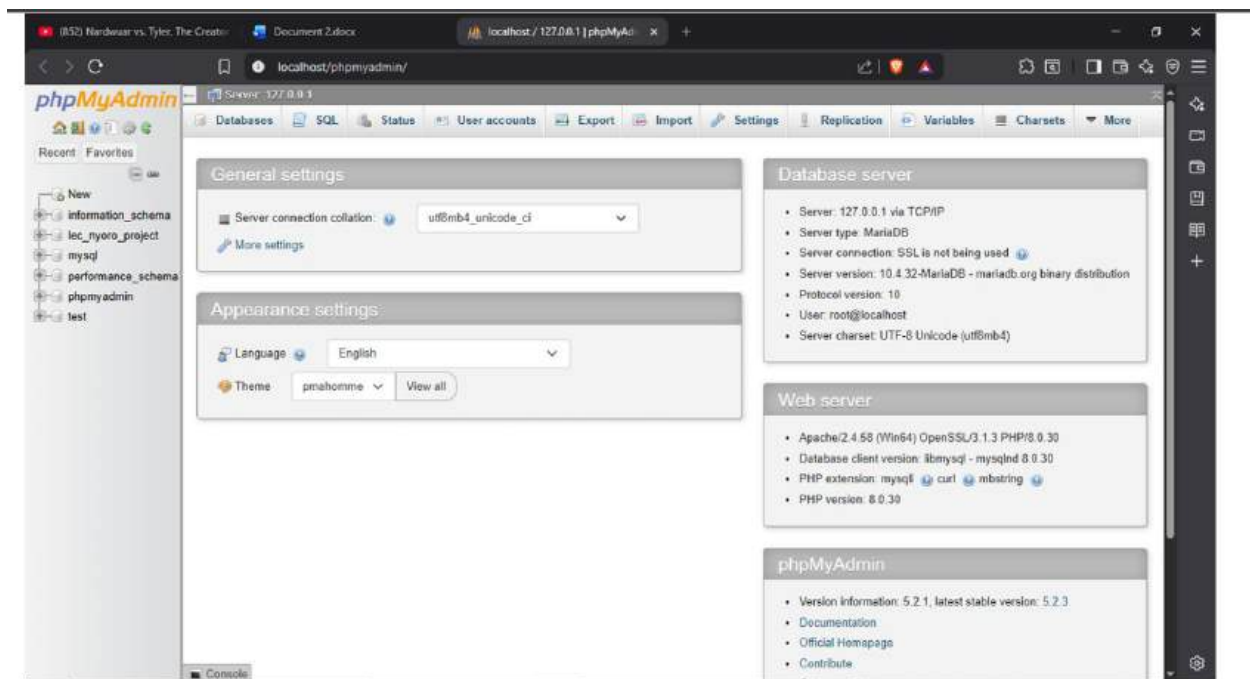
Github link; <https://github.com/wendykingori61/nchi-supermarket>

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WEEK 1; XAMPP LOCAL ENVIRONMENT SETUP





REFLECTION

Mastered local server architecture deployment by configuring XAMPP (Apache and MySQL).
Learned the difference between client-side and server side execution and organized a modular,
professional project directory structure for proper software development workflow.

WEEK 2; UI/UX AND DESIGN FOUNDATIONS

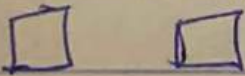
Layout Structure

Header

Nchi Supermarket

Cart (0)

Products Grid



Shopping Cart

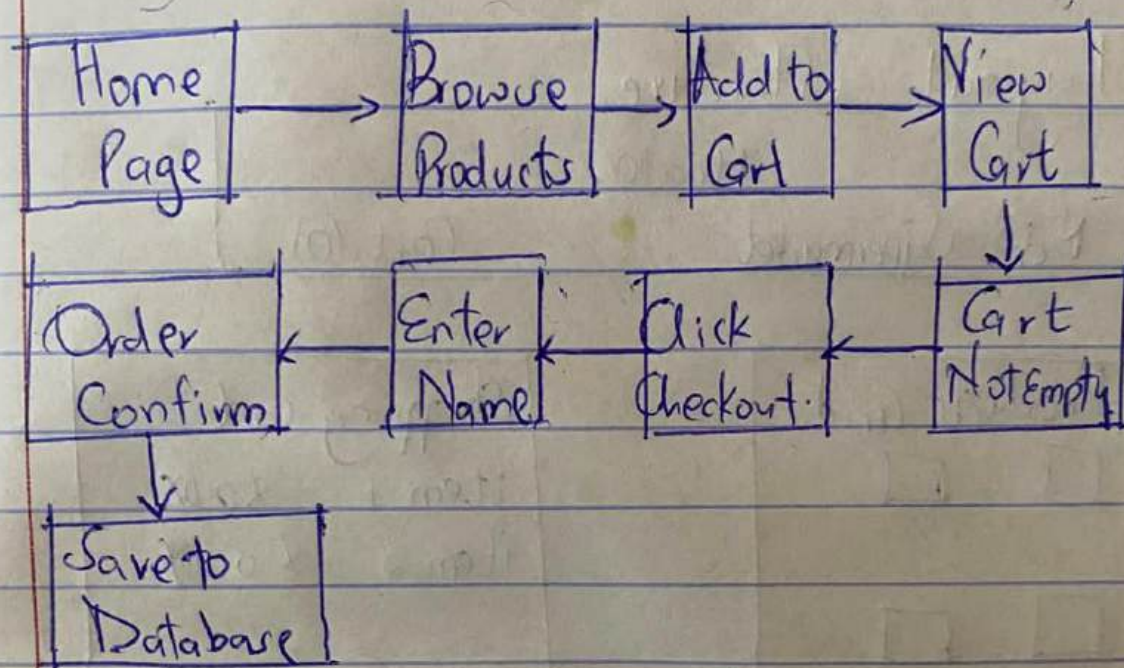
item 1 \$0.0X

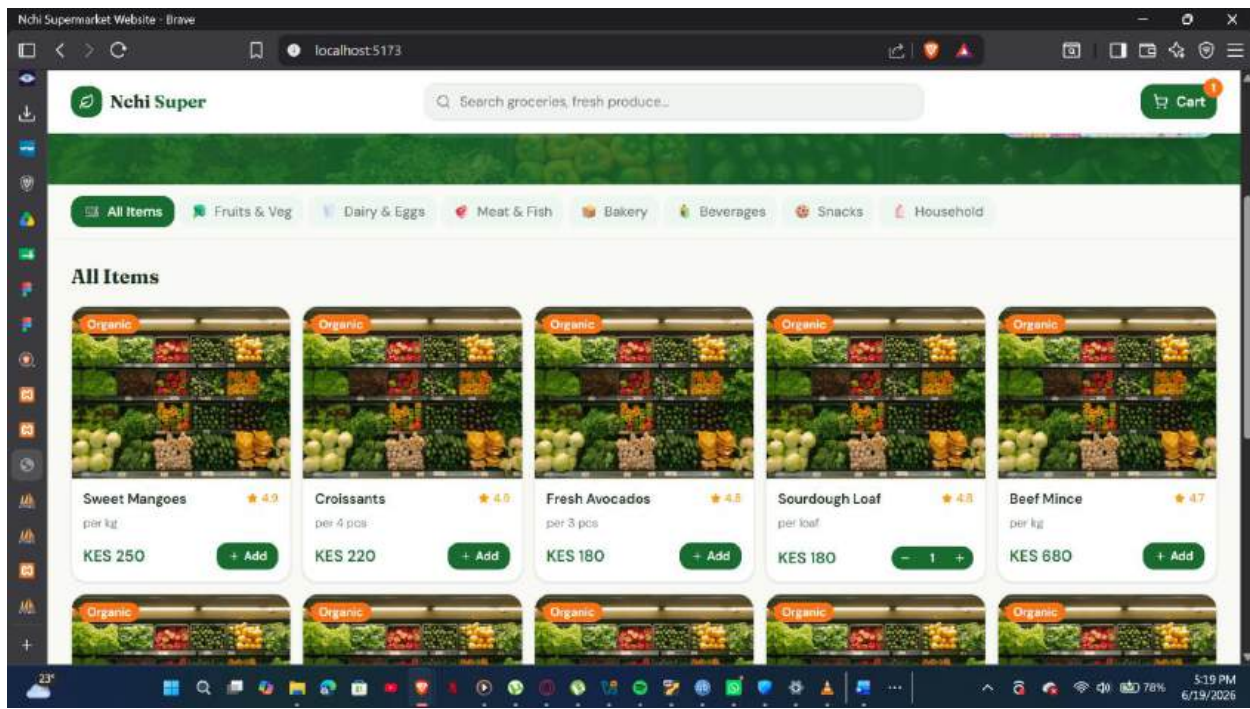
item 2 \$0.0X

Total: \$XX.XX

[checkout]

USER FLOW

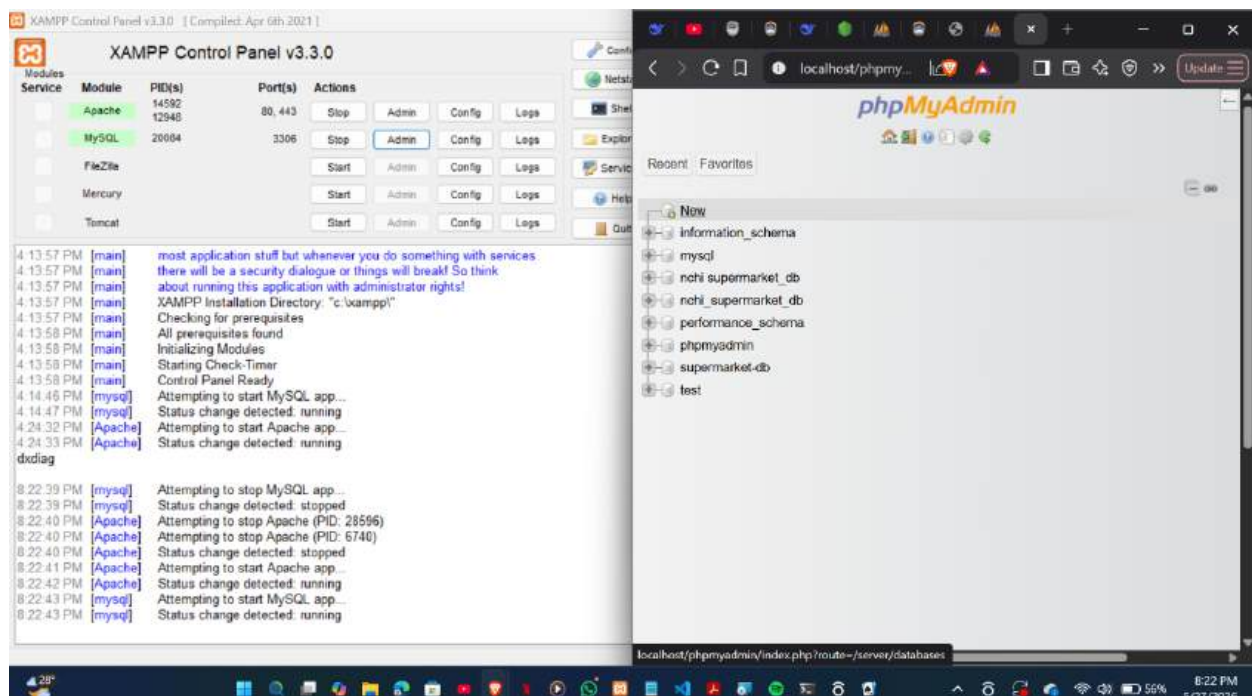


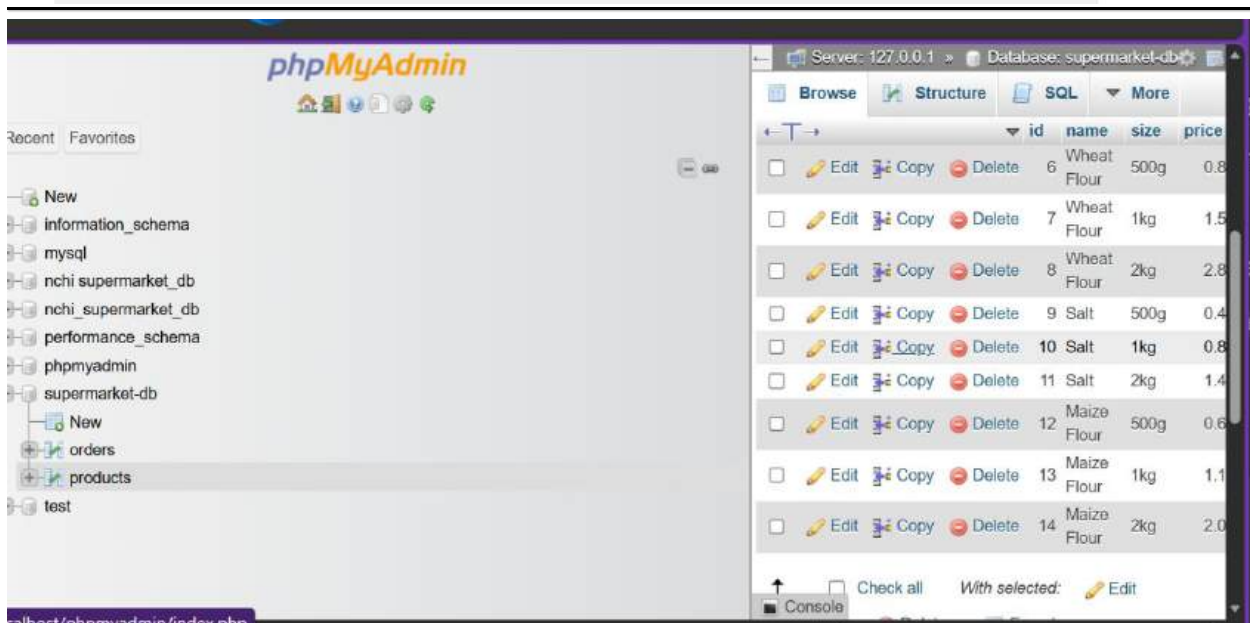


REVIEW

Nchi Supermarket offers a clean, intuitive interface with a green-themed design that feels fresh and trustworthy. The two-column layout keeps products visible while the cart stays accessible on the right, and hover effects on product cards provide satisfying visual feedback. All actions trigger clear success or error messages, making the shopping experience smooth and user-friendly.

WEEK 3;FRONTEND SCRIPTING AND BACKEND PROGRAMMING FOUNDATIONS





REVIEW

Nchi Supermarket is a full-stack e-commerce website that sells milk, sugar, wheat flour, salt, and maize flour in multiple sizes. It uses **Node.js + Express** on the backend, **MySQL** for the database (managed via phpMyAdmin), and **HTML/CSS/JavaScript** on the frontend with **localStorage** for the cart (no cookies). The website connects to a database via a connection string, displays all products dynamically, and saves customer orders permanently.

WEEK 4: SERVER-SIDE COMPONENTS AND BACKEND DEVELOPMENT FOUNDATIONS

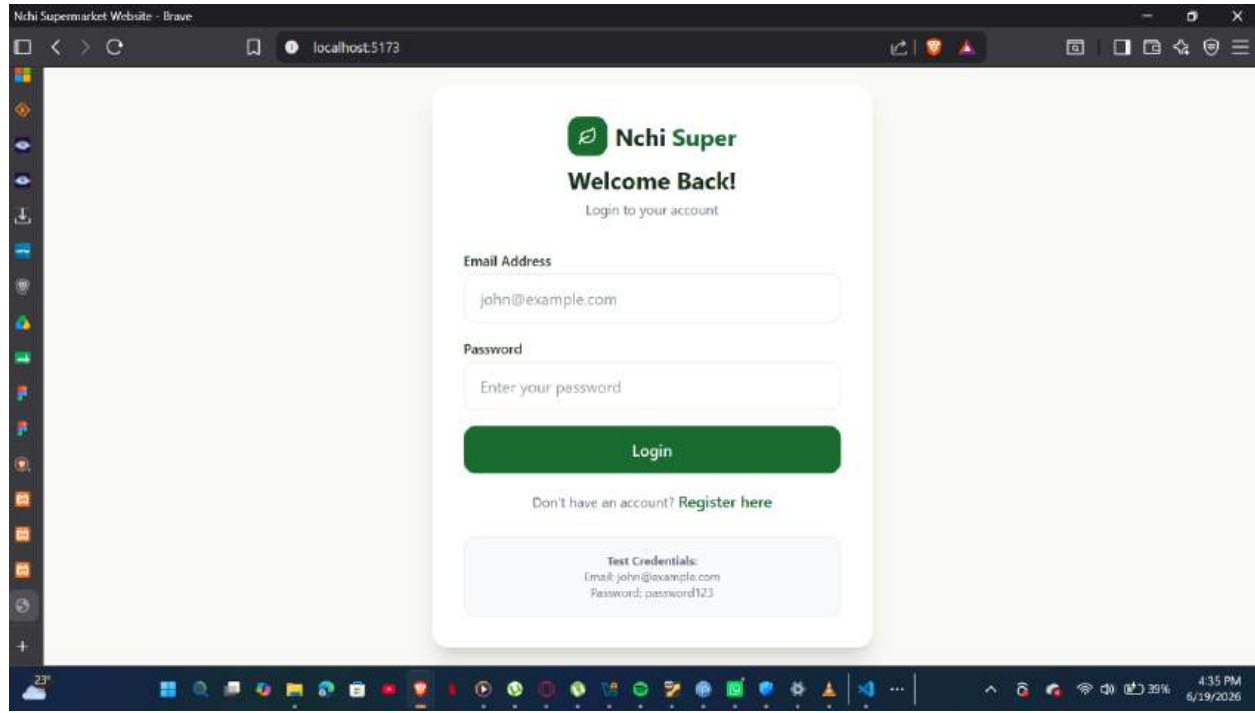


Fig 1 login form

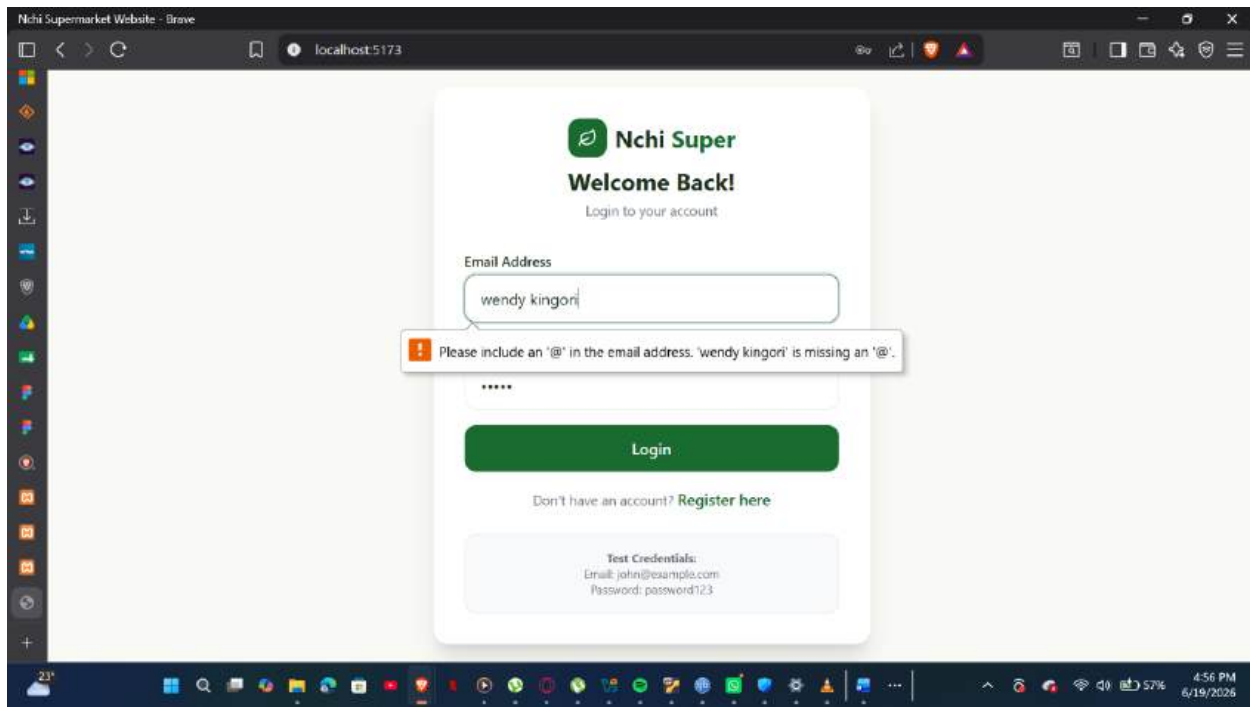


Fig 2 form validation

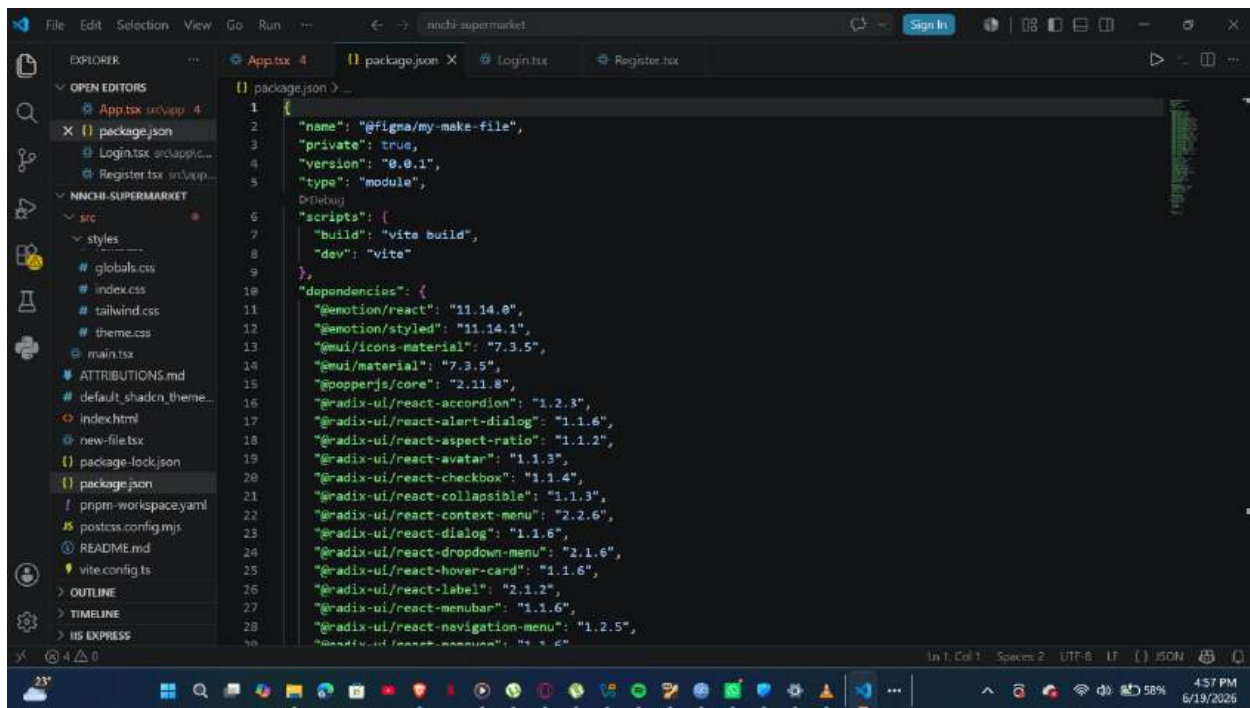


Fig 3 php processing

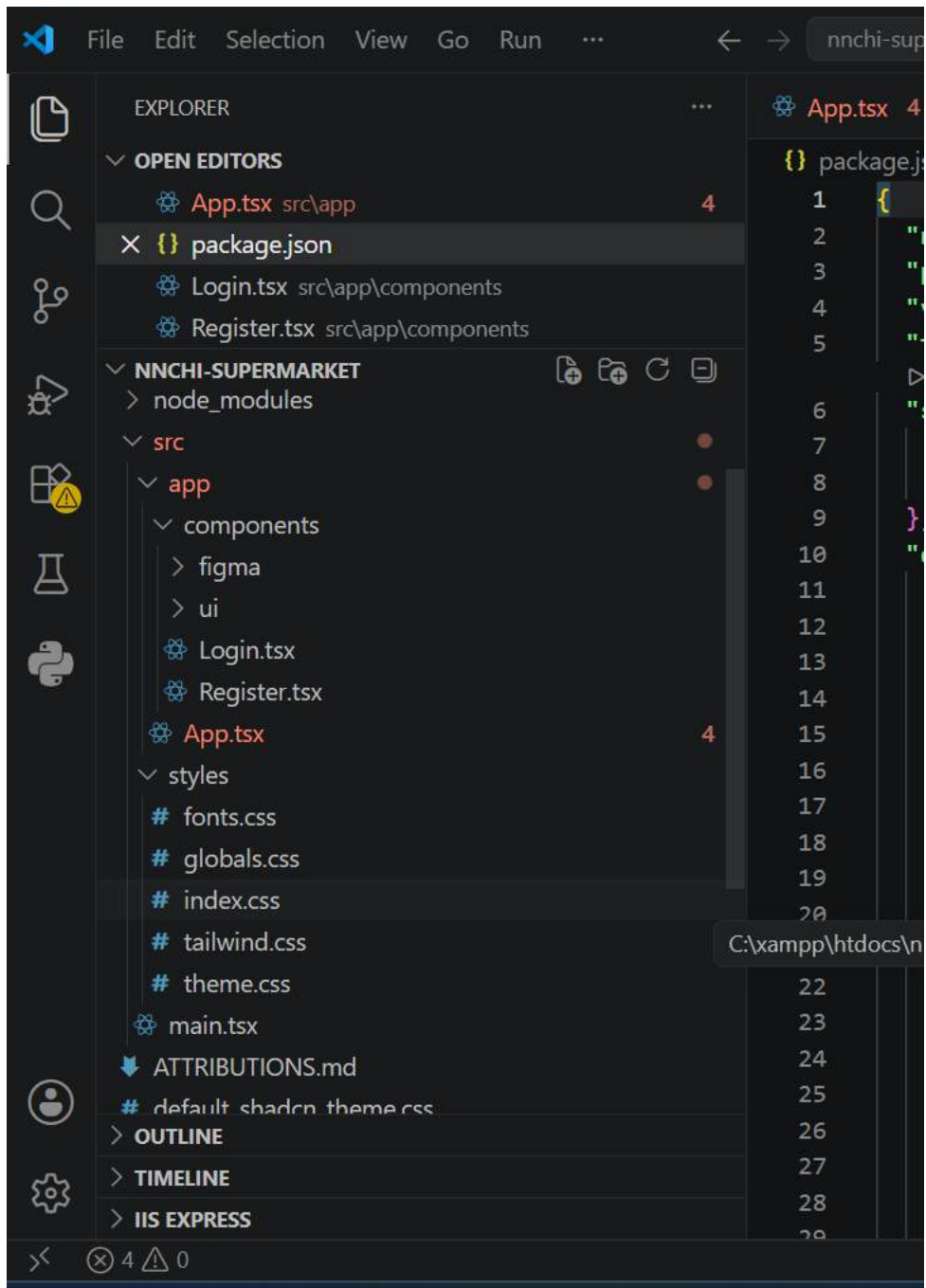


Fig 4 folder structure

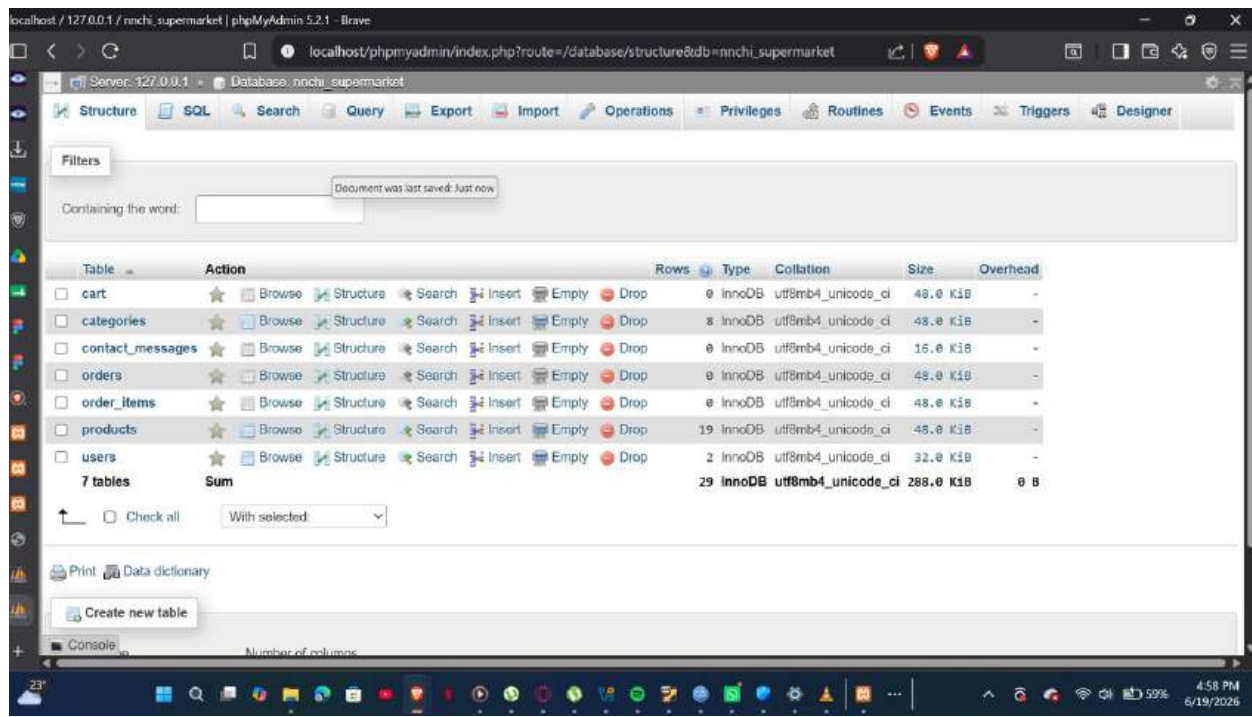


Fig 5 database connection

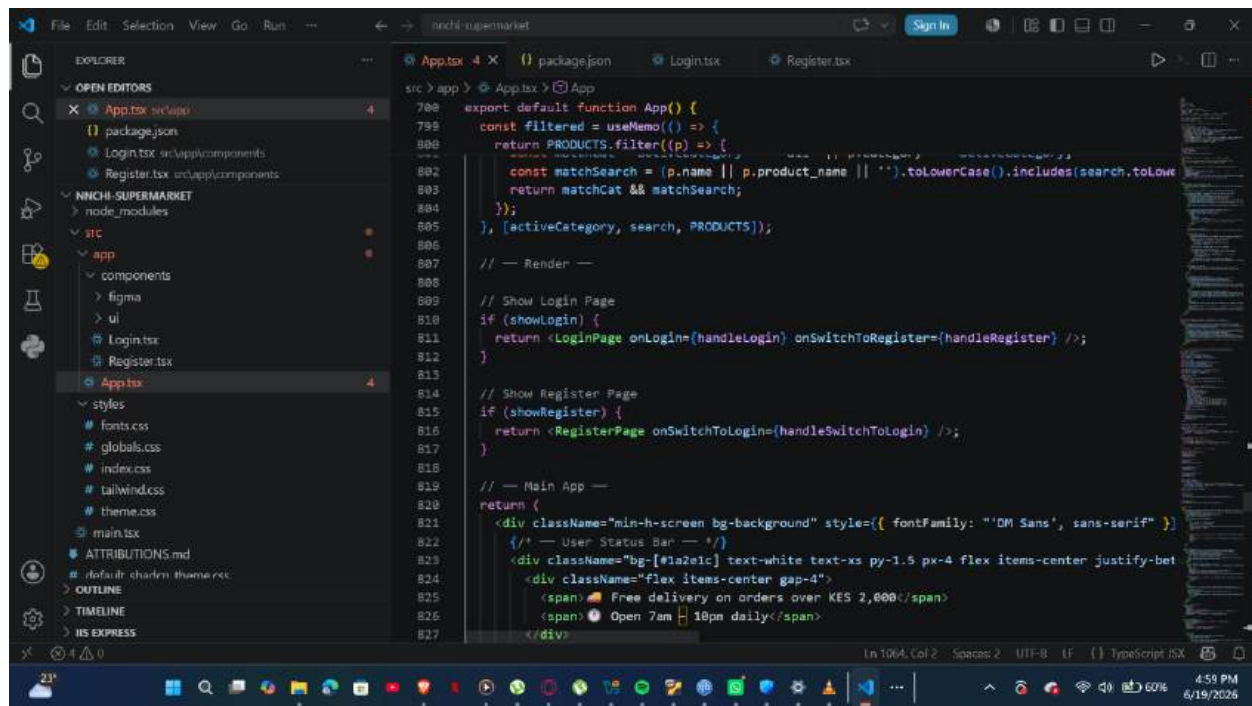


Fig 6 code snippets

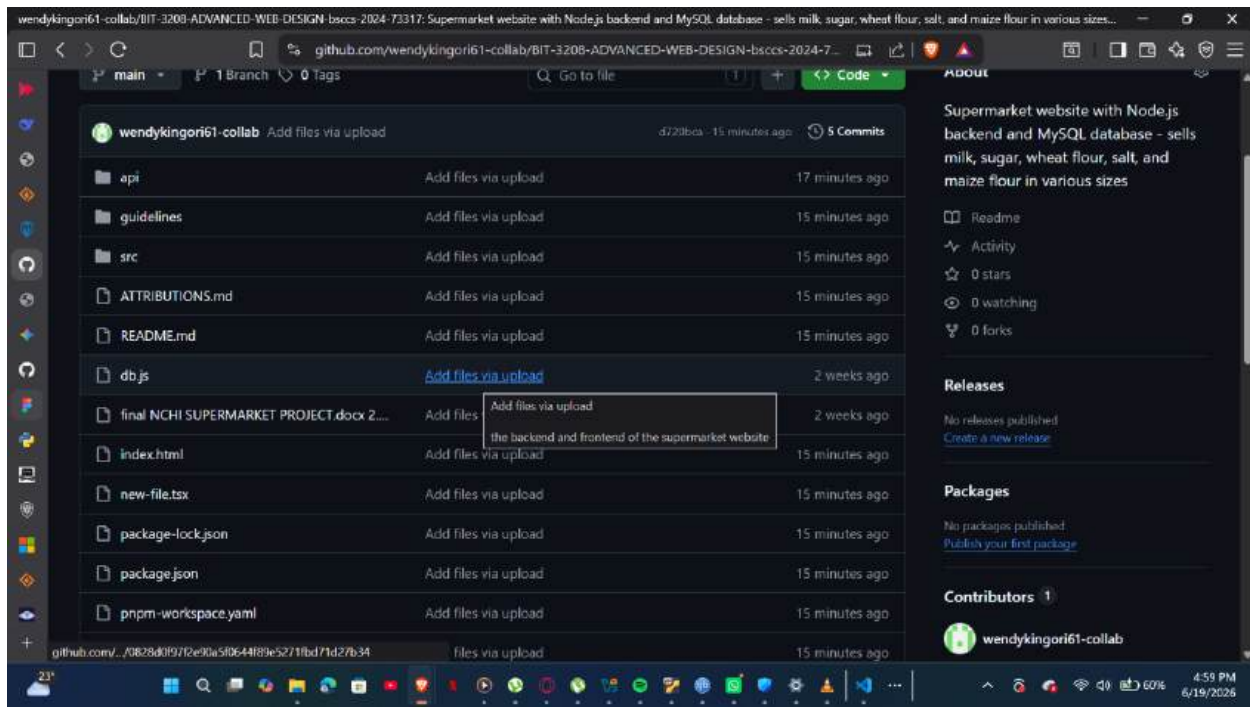


Fig 7 github updates

REVIEW

I successfully created a complete PHP backend API with database connection, authentication (login/register with password hashing), and product data endpoints for my Nnchi Supermarket e-commerce project.

WEEK 5; DATABASE COMPONENTS AND CRUD OPERATIONS

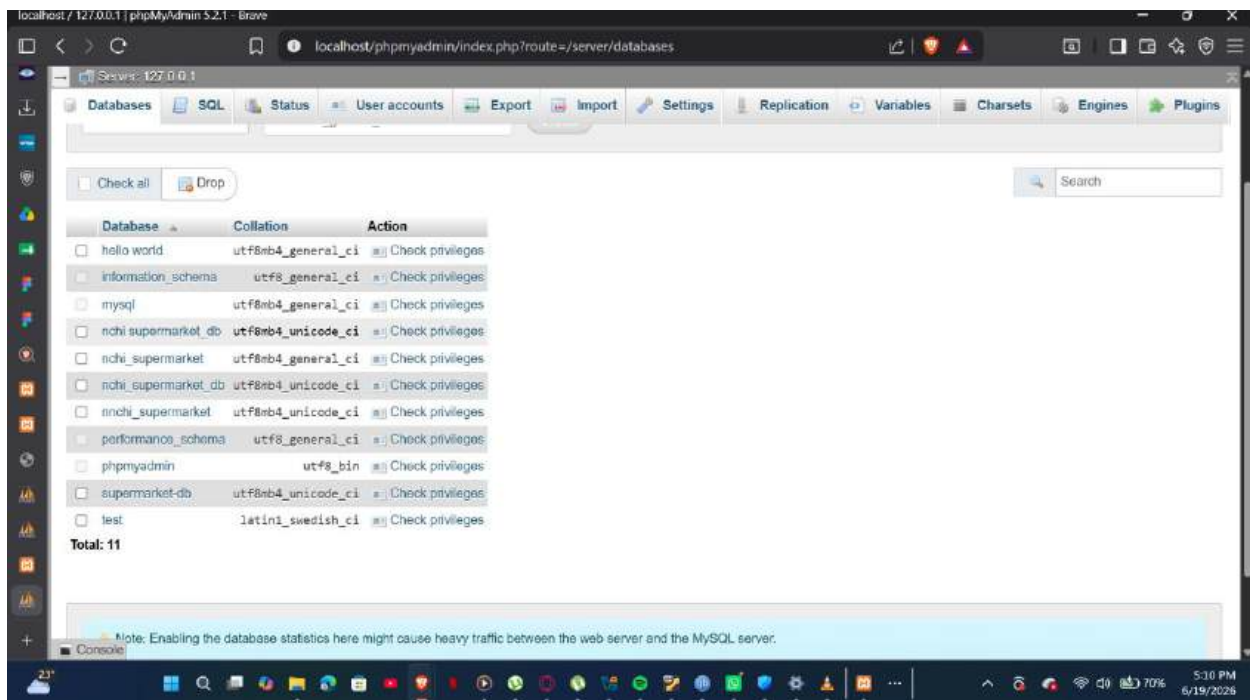


Fig 1 database creation

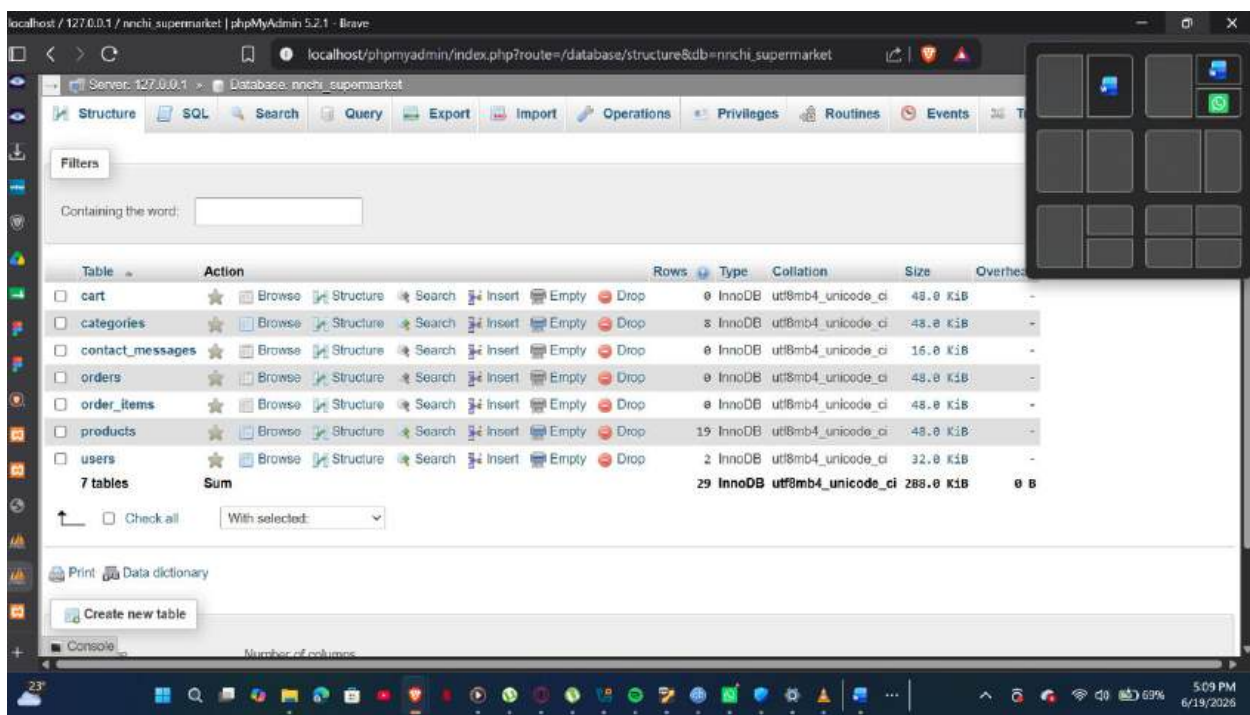


Fig 2 table creation

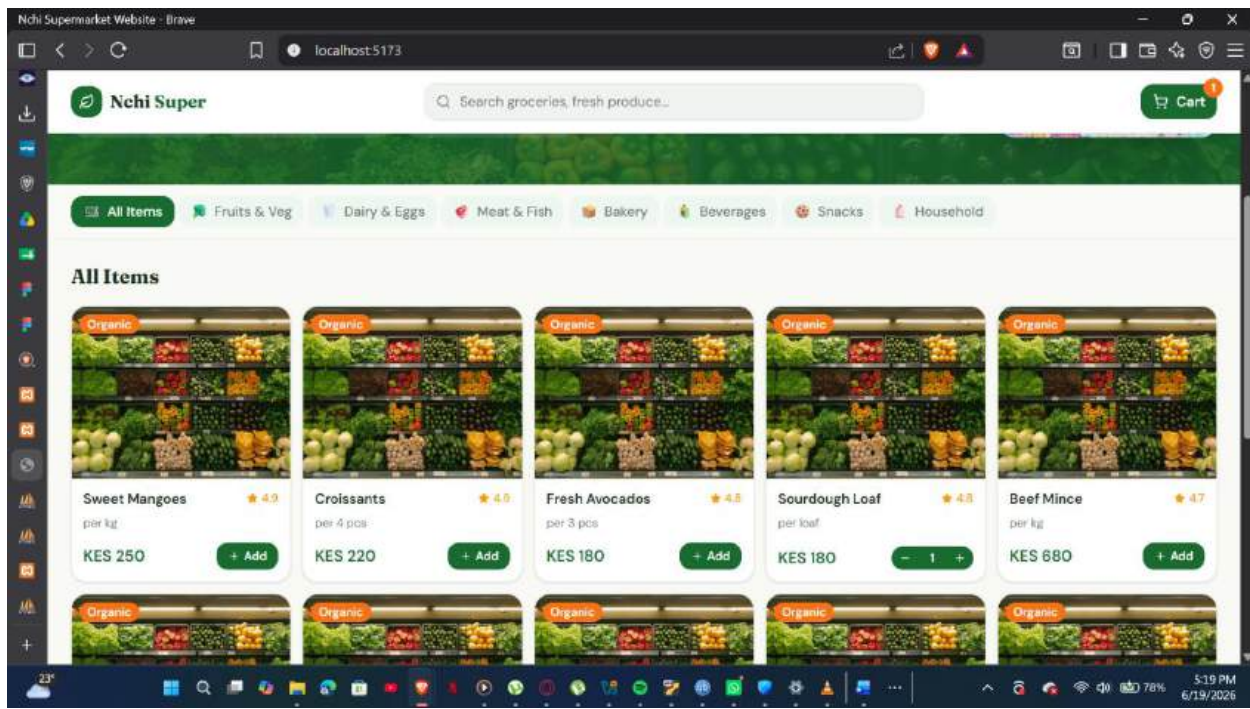


Fig 4 create

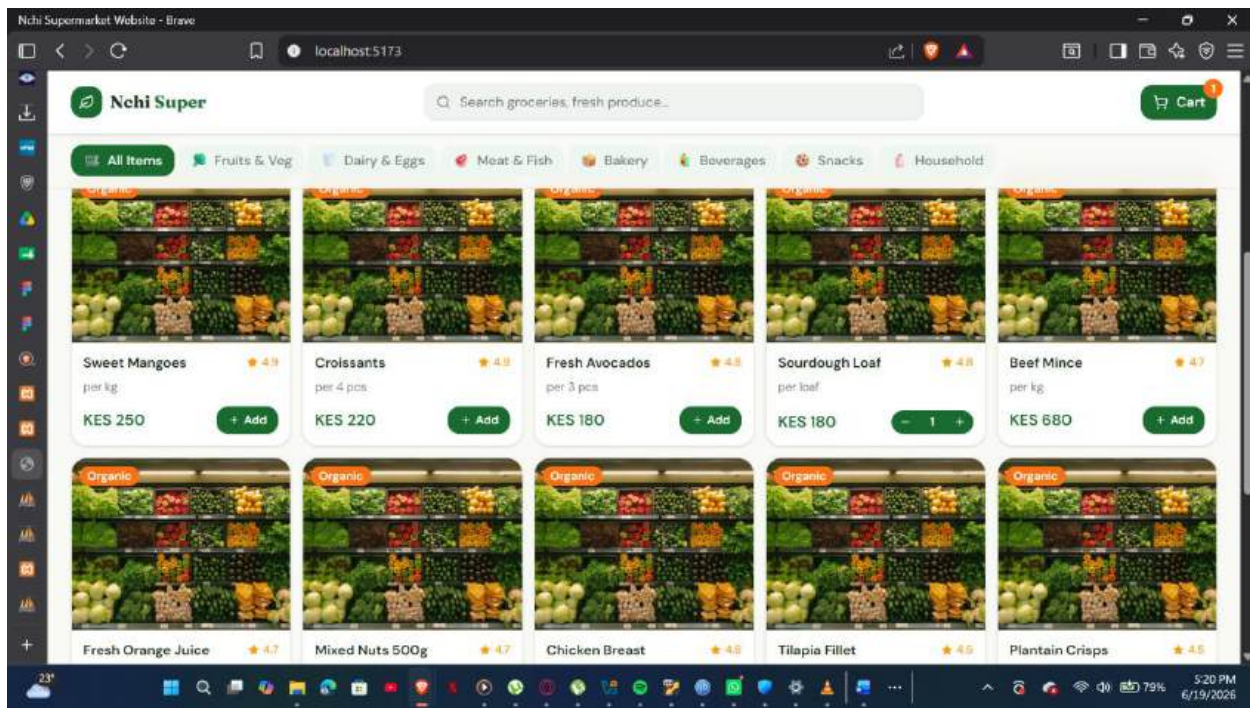


Fig 5 read

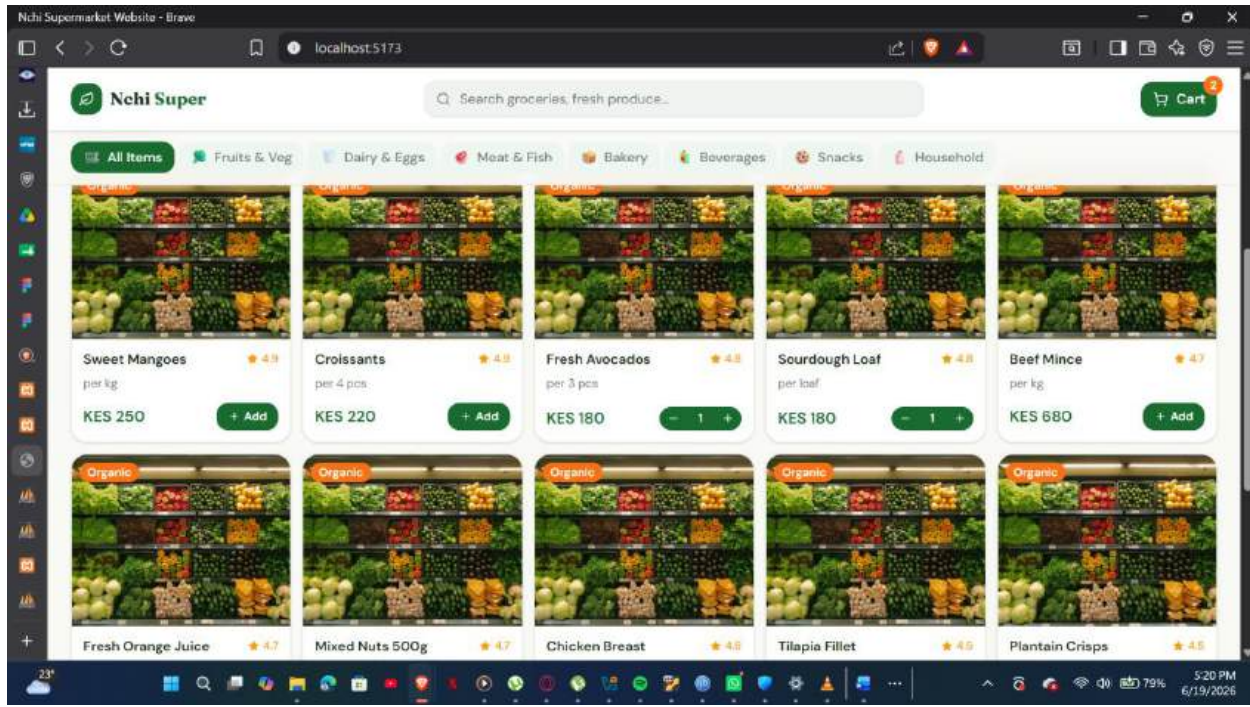


Fig 6 update

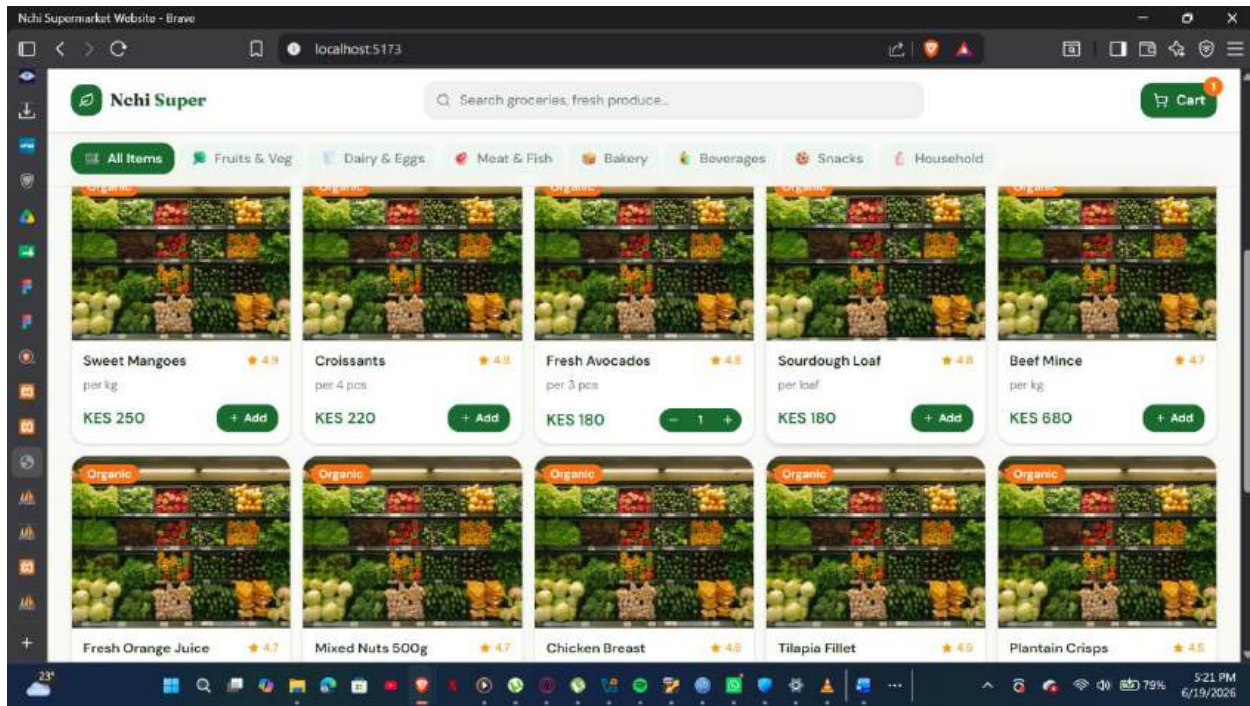


Fig 7 delete

Review

Successfully created MySQL database with tables for products, categories, and users.

Implemented CRUD operations where users can view products, add to cart, update quantities, and remove items. Connected PHP backend to MySQL database.

WEEK 6; DATABASE INTEGRATION AND CRUD OPERATIONS

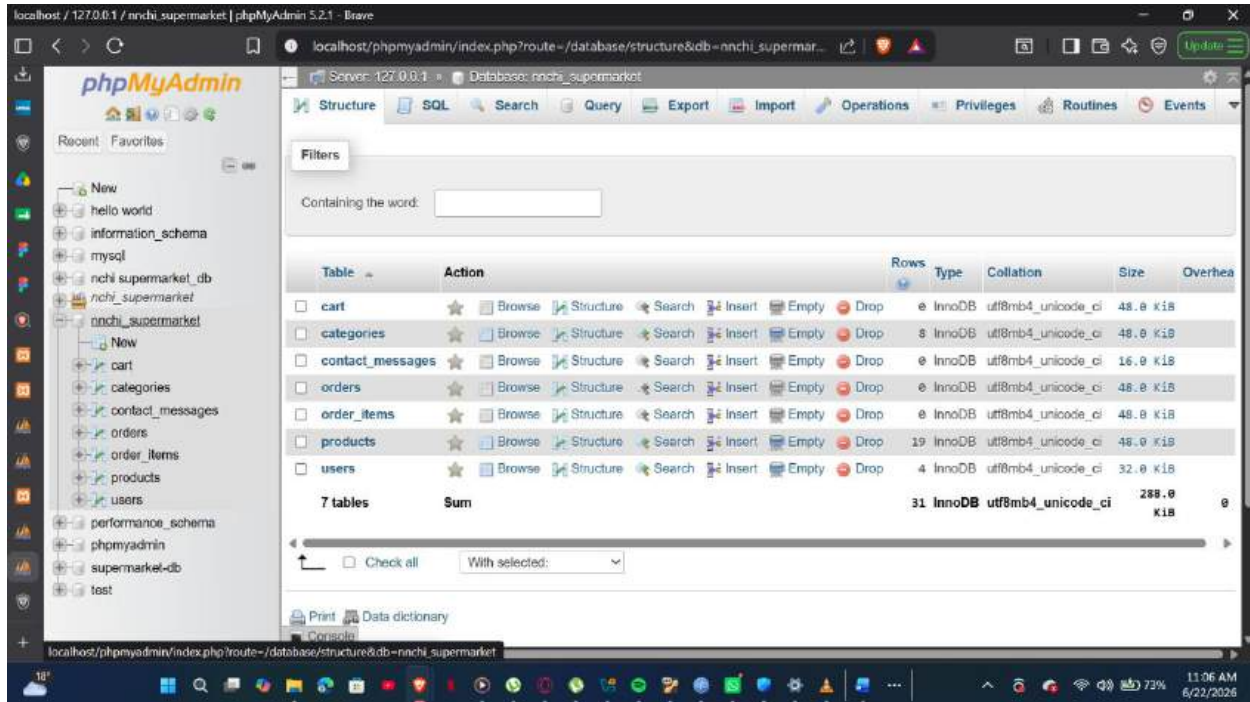


Fig 1 database

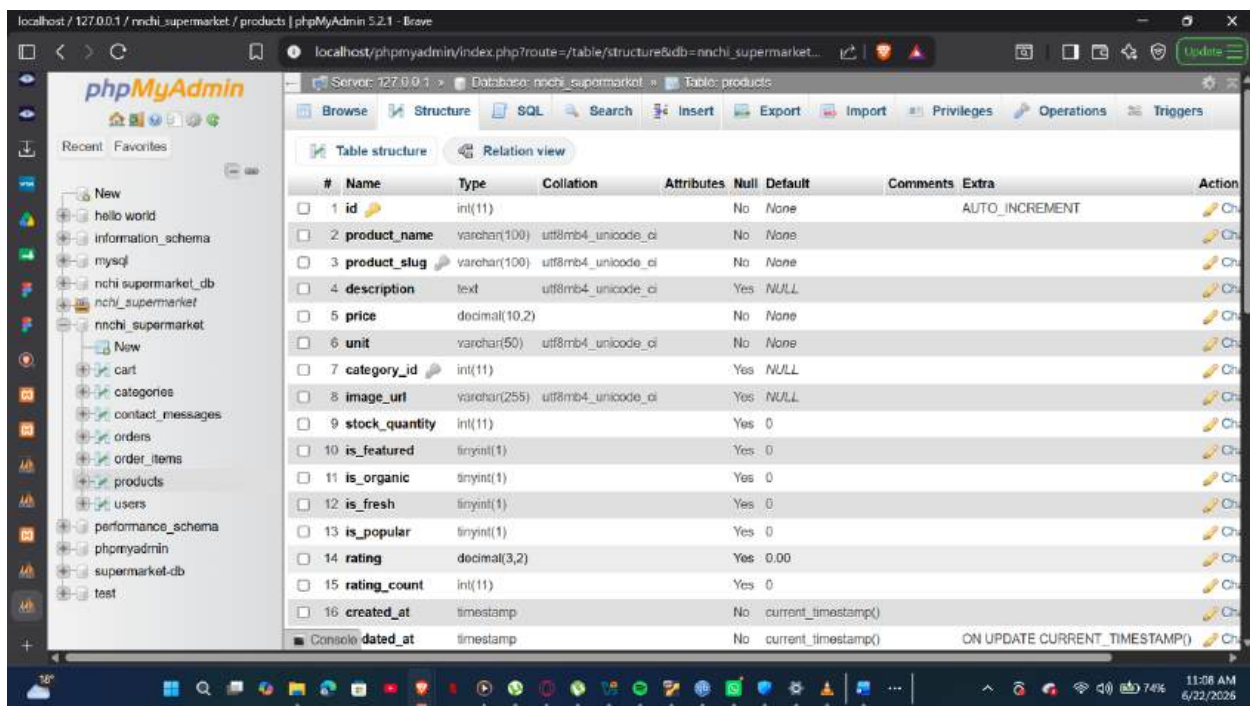


Fig 2 products table

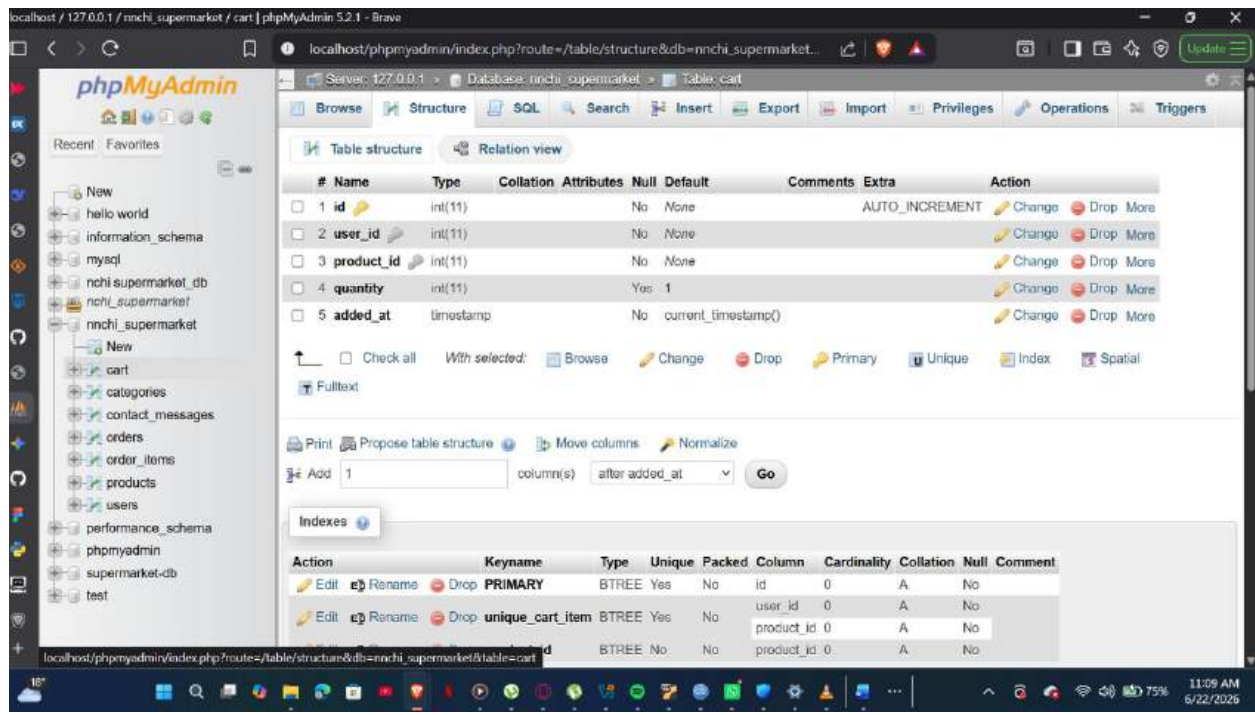


Fig 3cart table

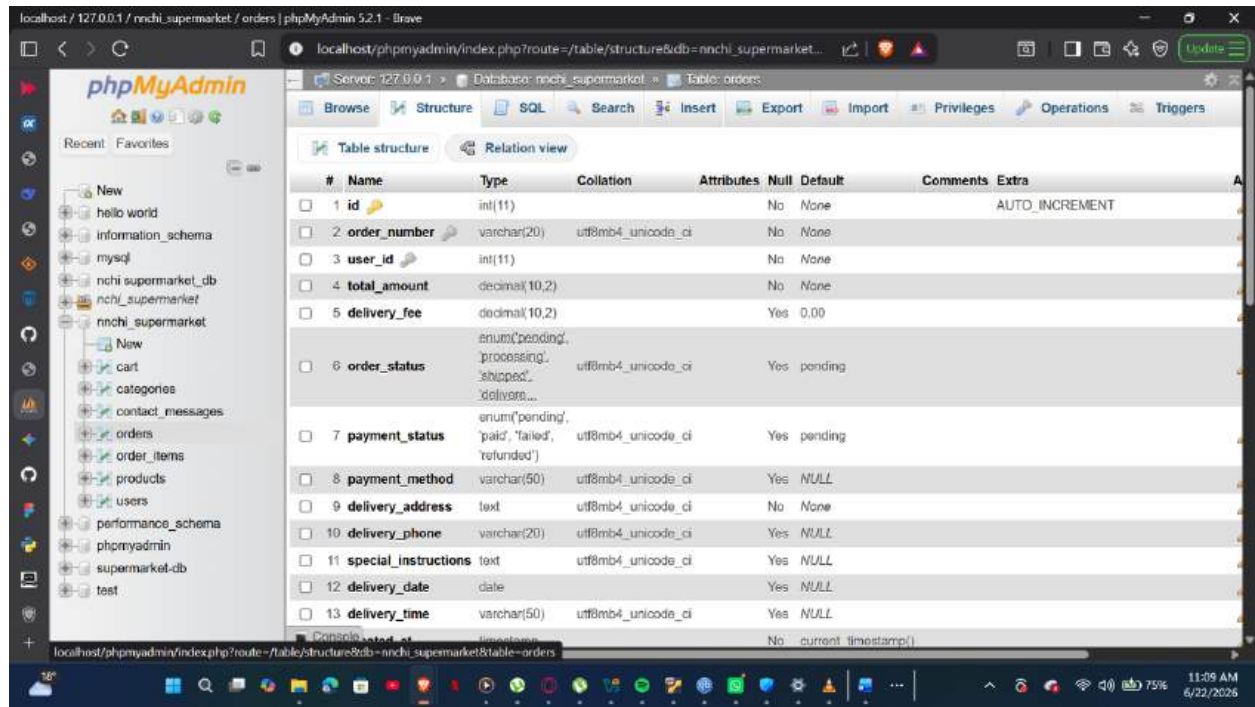


Fig 4 orders table

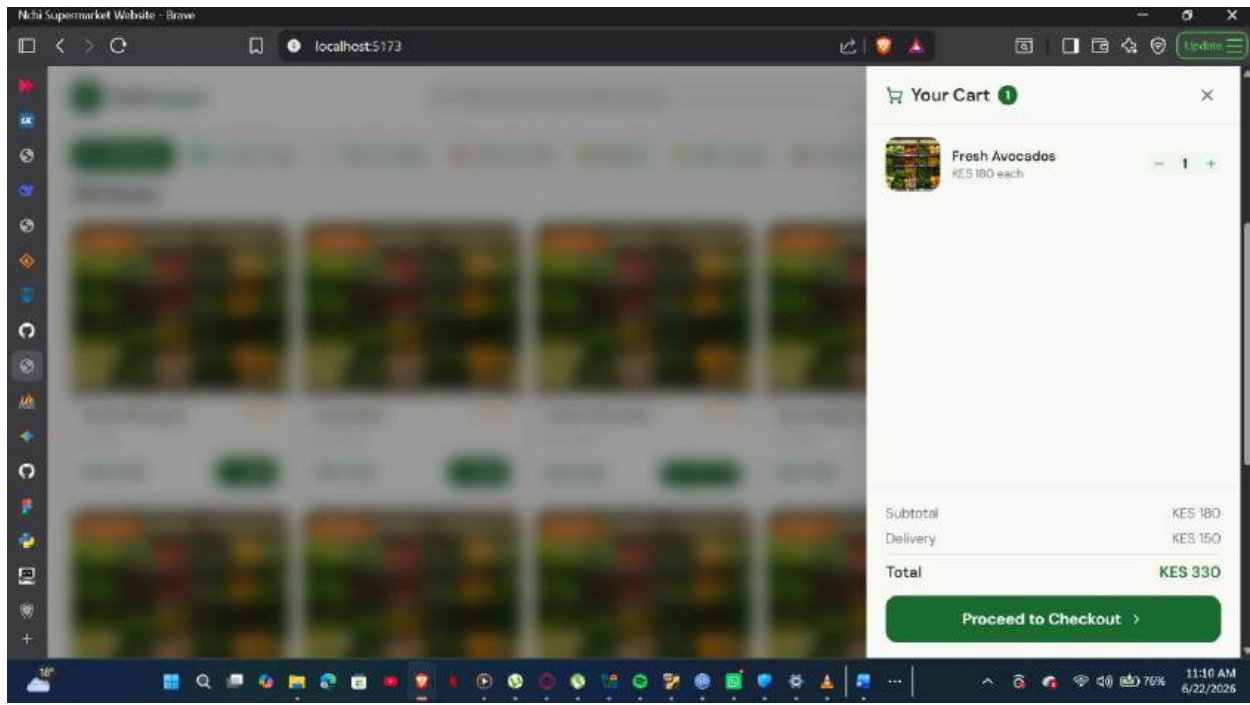


Fig 5 create

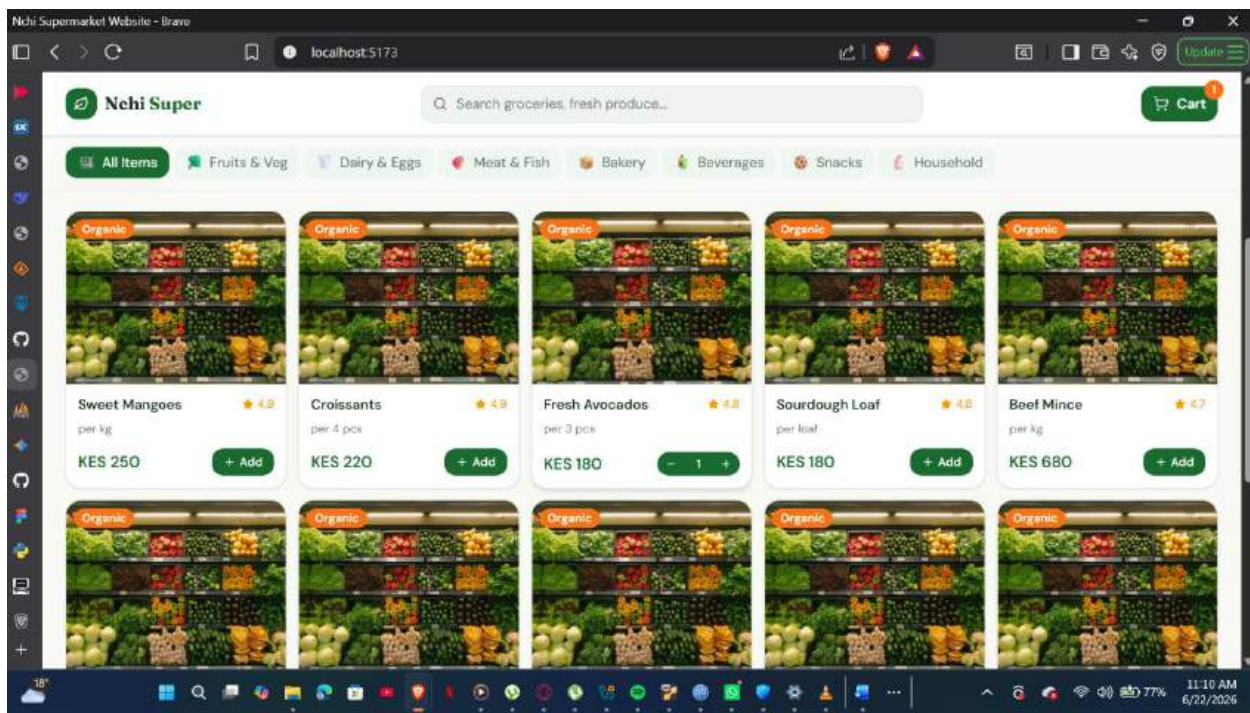
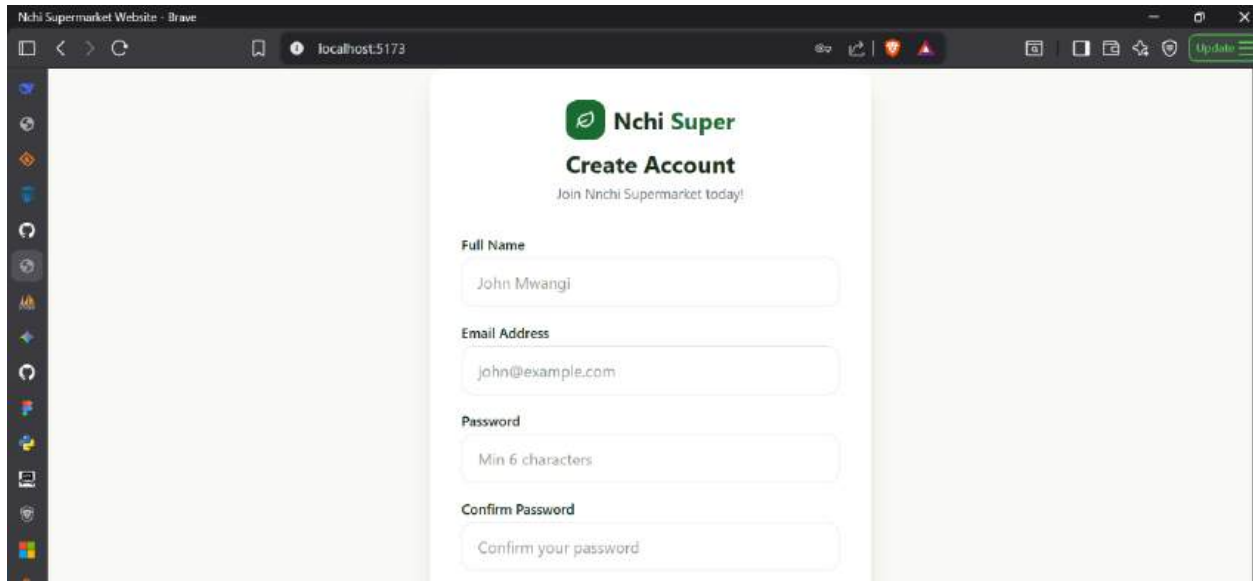


Fig 6 read

REVIEW

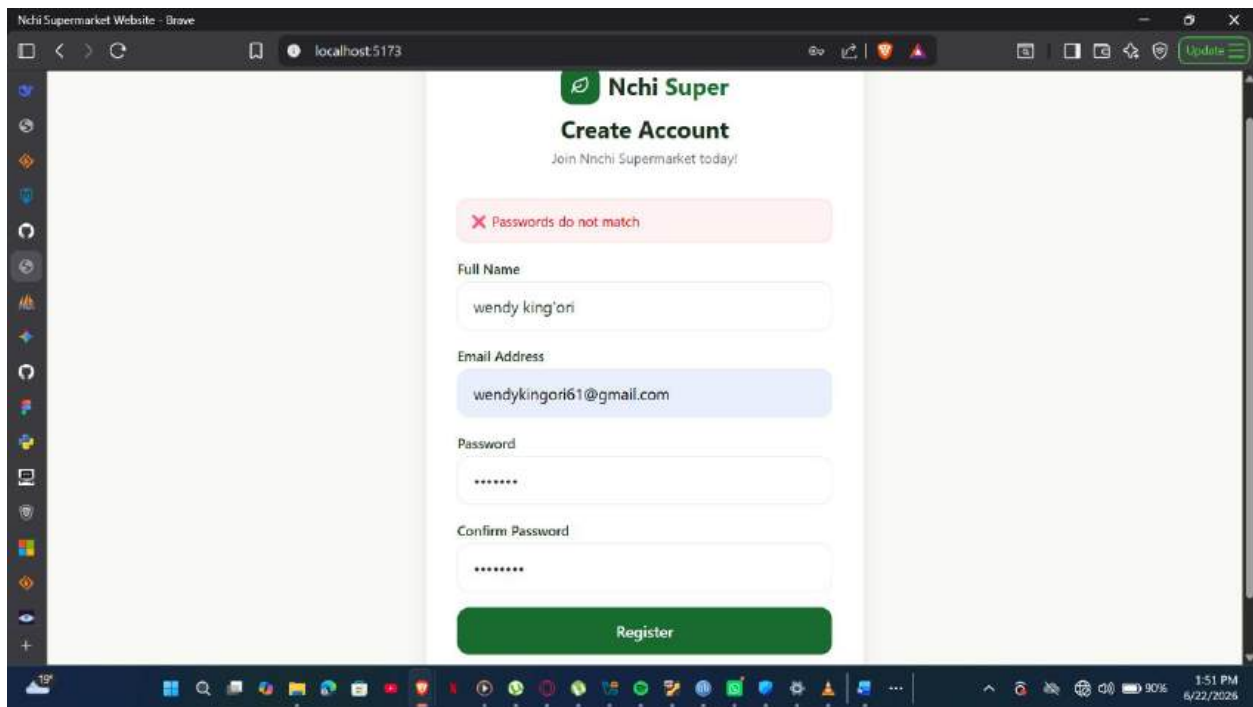
Integrated database with shopping cart and checkout system. Users can add multiple products, update quantities, remove items, and place orders. Order confirmation generates unique order numbers. SQL queries validated database operations.

WEEK 7: USER AUTHENTICATION AND SESSION MANAGEMENT



The screenshot shows a web browser window titled "Nchi Supermarket Website - Brave" with the address bar displaying "localhost:5173". The page content features the "Nchi Super" logo and the heading "Create Account" with the subtext "Join Nnchi Supermarket today!". Below this, there are four input fields: "Full Name" (containing "John Mwangi"), "Email Address" (containing "john@example.com"), "Password" (with a placeholder "Min 6 characters"), and "Confirm Password" (with a placeholder "Confirm your password").

Fig 1 registration form



This screenshot shows the same "Create Account" form as Figure 1, but with updated data and a validation error. The "Full Name" field now contains "wendy king'ori", the "Email Address" field contains "wendykingori61@gmail.com", and both the "Password" and "Confirm Password" fields are filled with "*****". A red error message "X Passwords do not match" is displayed above the "Full Name" field. A green "Register" button is located at the bottom of the form. The browser window title is "Nchi Supermarket Website - Brave" and the address bar shows "localhost:5173". The Windows taskbar at the bottom indicates the time is 1:51 PM on 6/22/2026.

Fig 2 registration validation

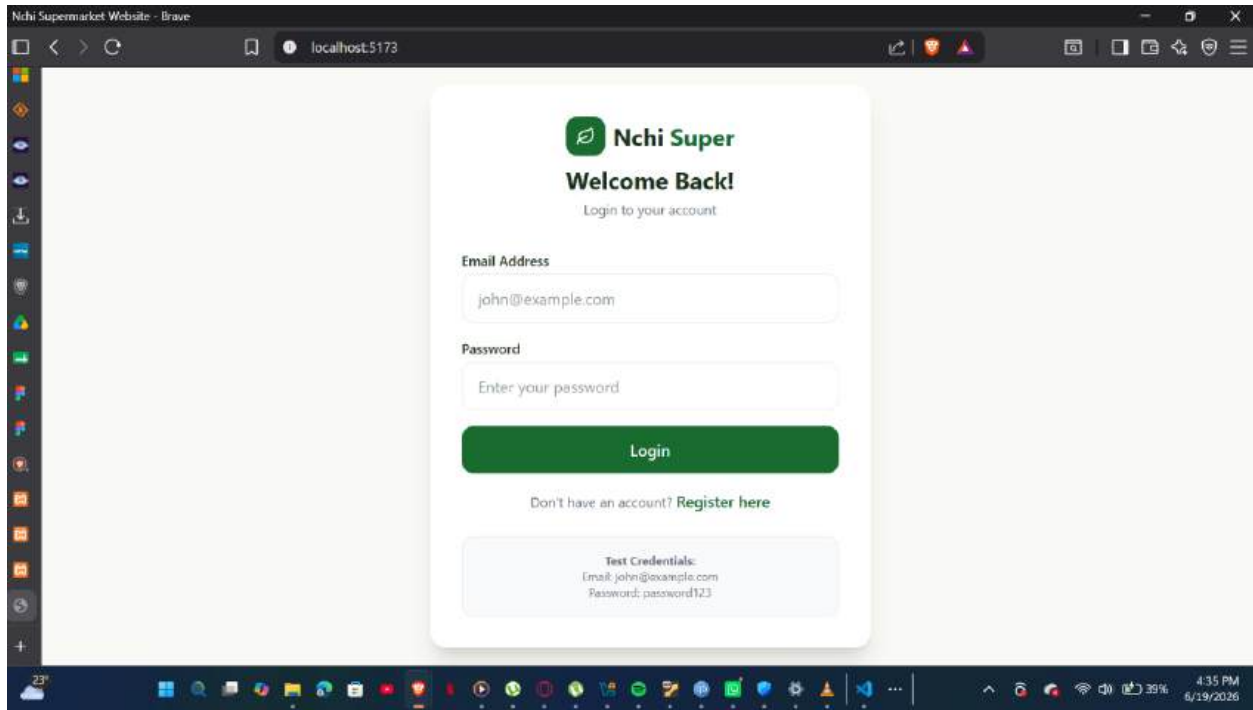


Fig3 login form

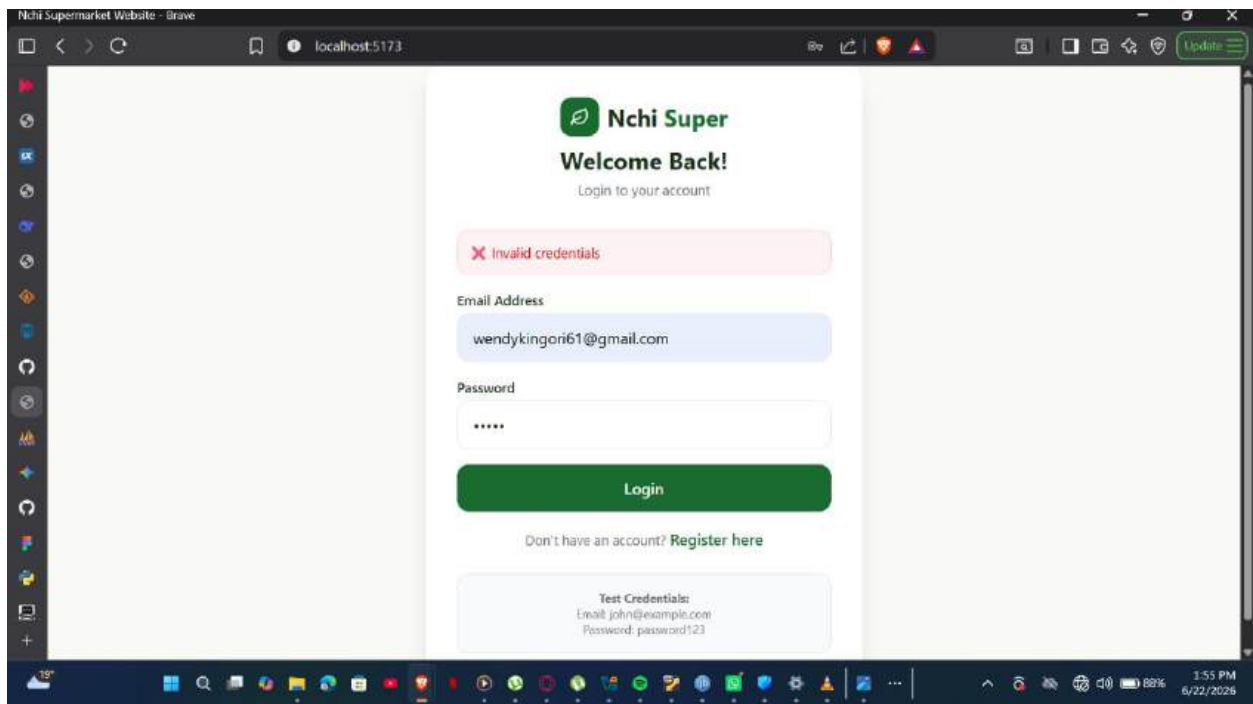


Fig 4 login error

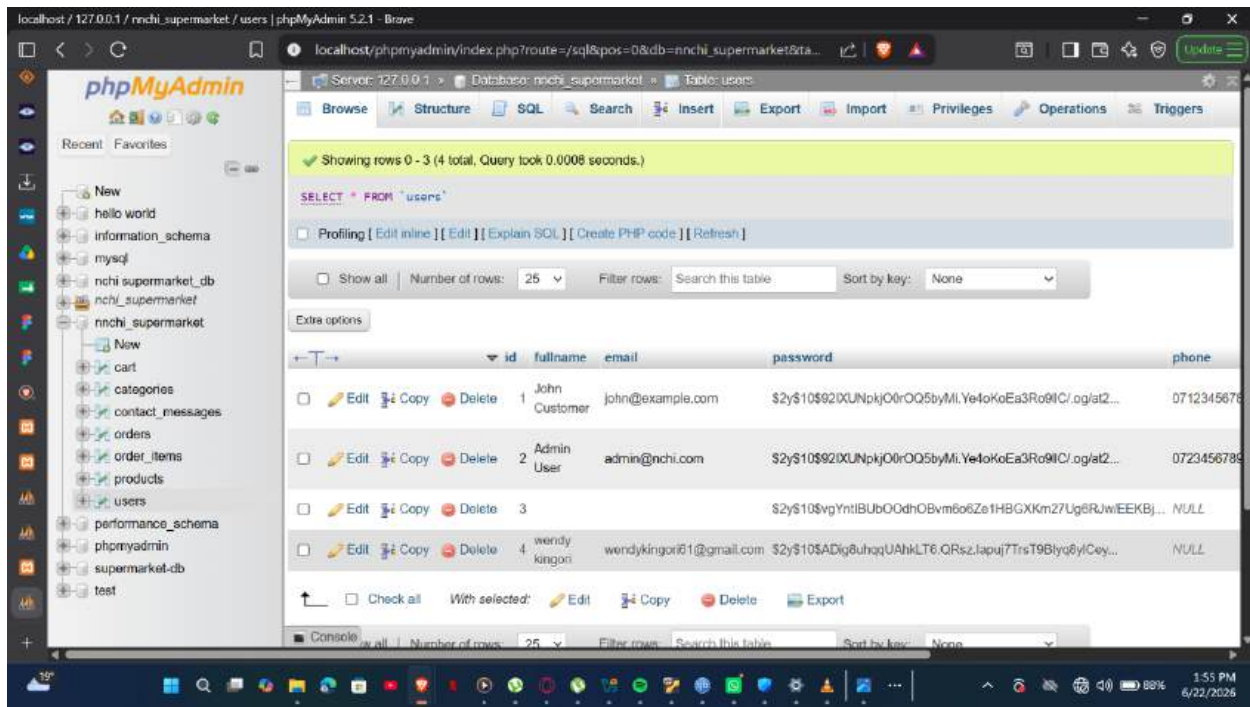


Fig 5 password hashing



Fig 6 successful login

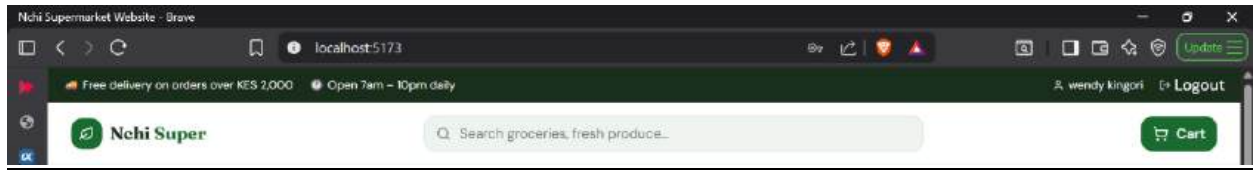


Fig 7 session active

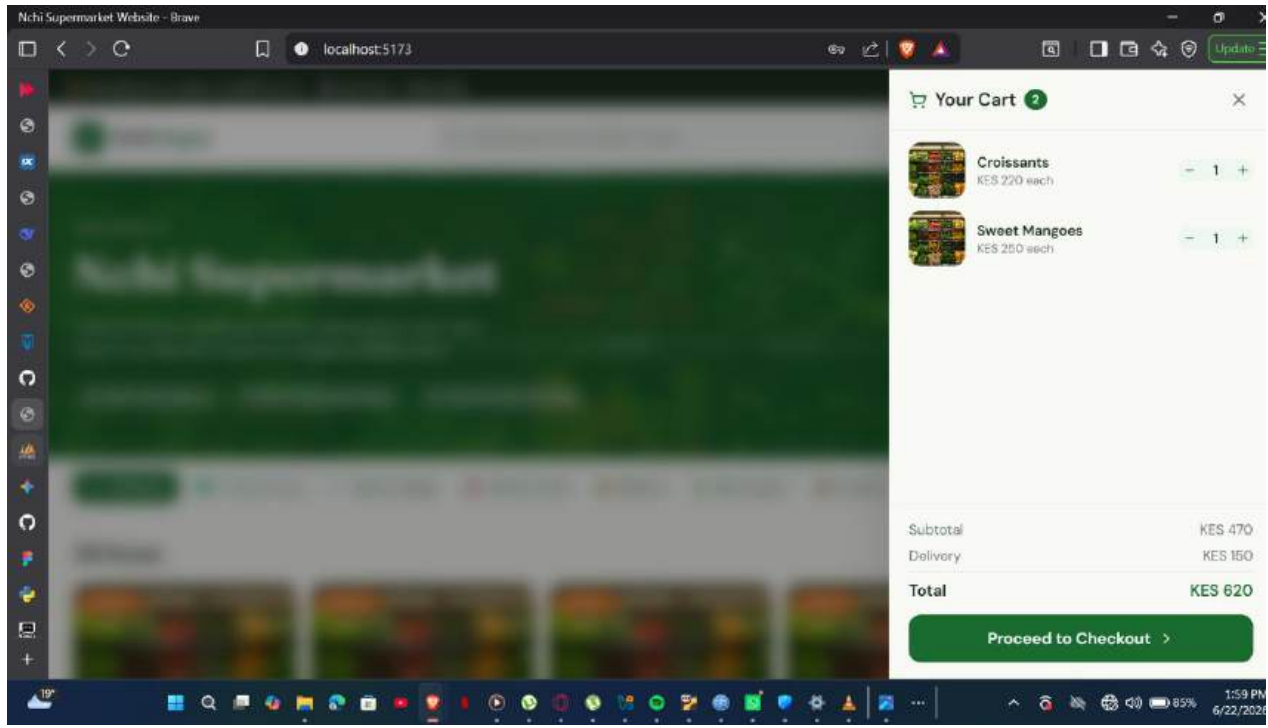


fig 8 protected page access



Fig 8 logout

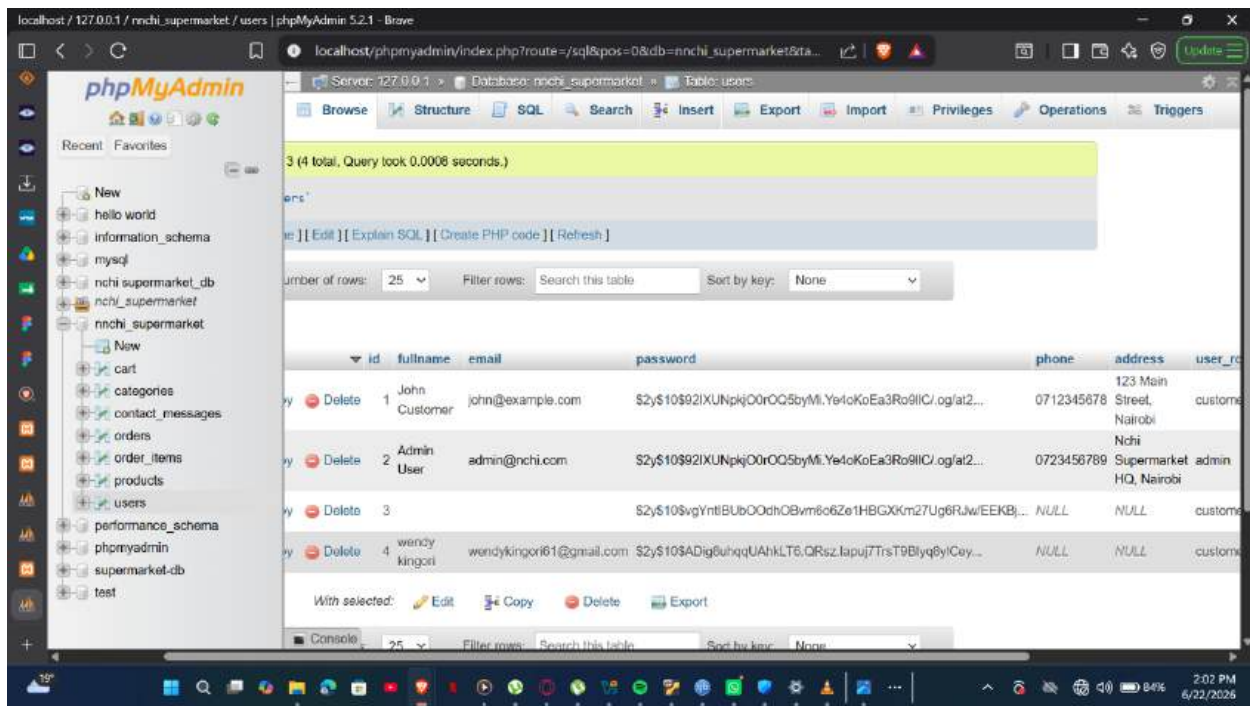


Fig 9 user data

Review

Implemented secure user authentication with registration, login, password hashing, and session management. Users can register accounts, login securely with hashed passwords, access protected pages, and logout. Sessions maintained using React state and localStorage.

WEEK 8; RESPONSIVE WEB DESIGN AND MOBILE-FIRST DEVELOPMENT

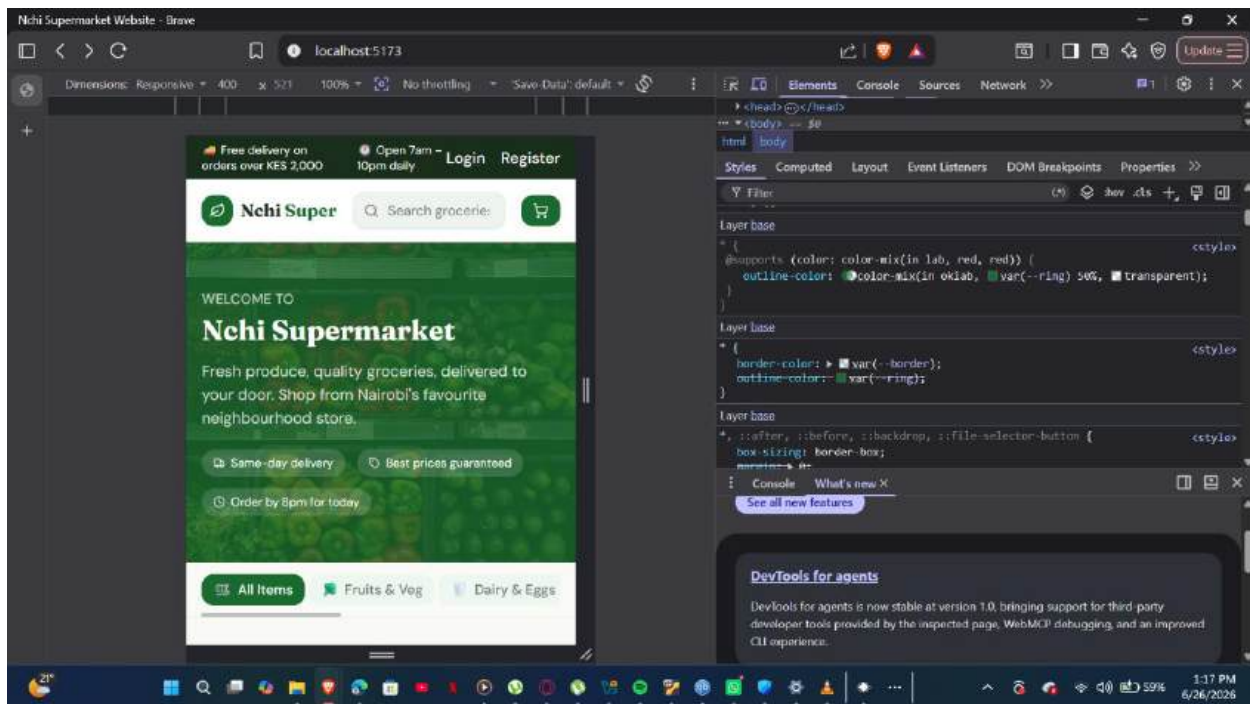


Fig 1 Mobile view

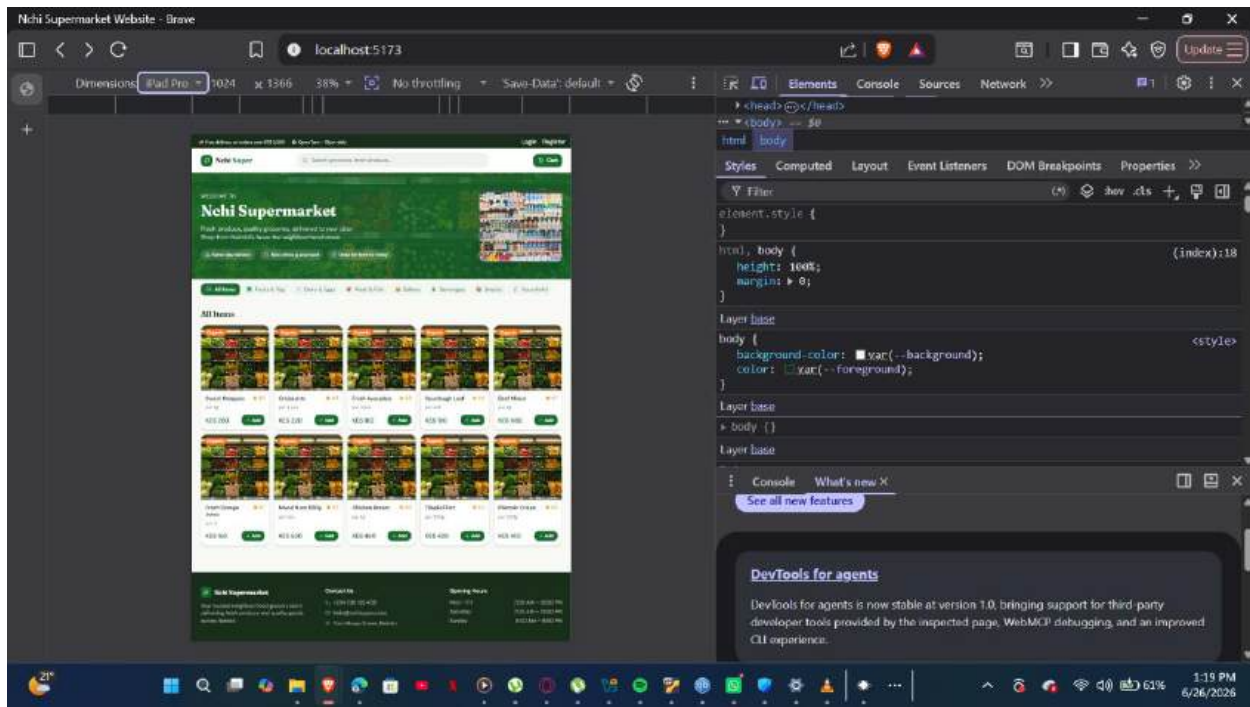


Fig 2 tablet view



Fig 3 desktop view

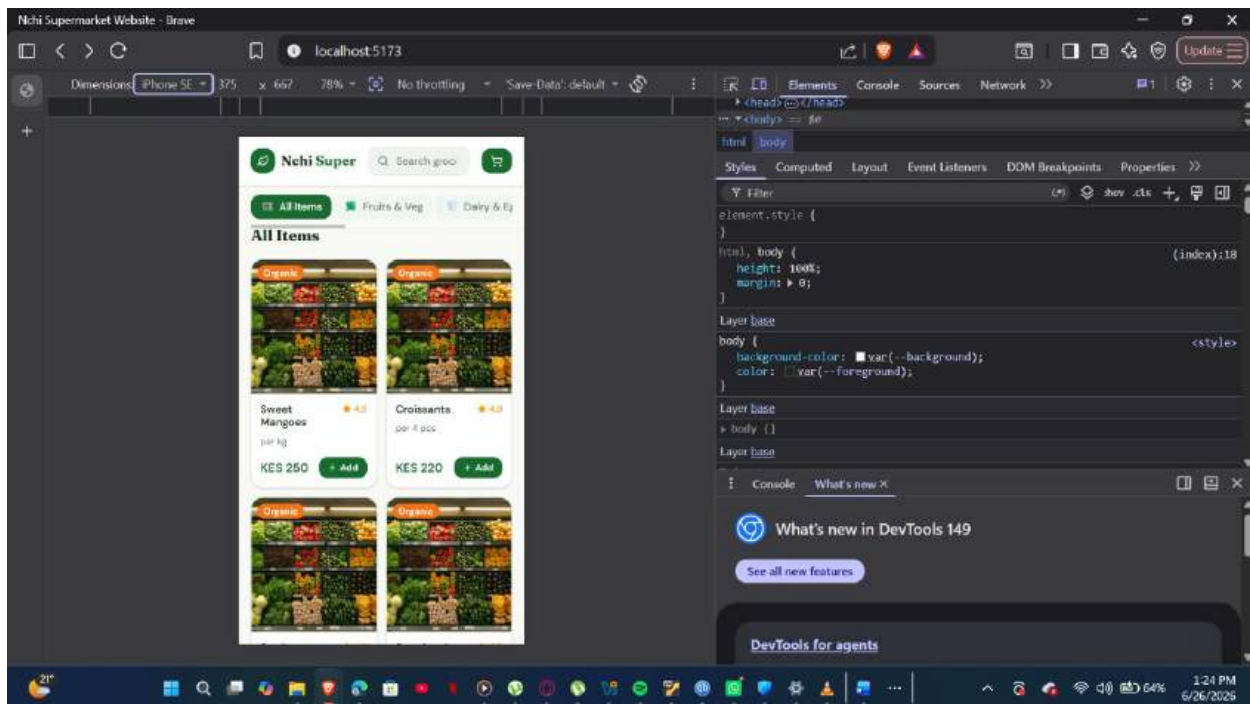


Fig 4 product_grid mobile

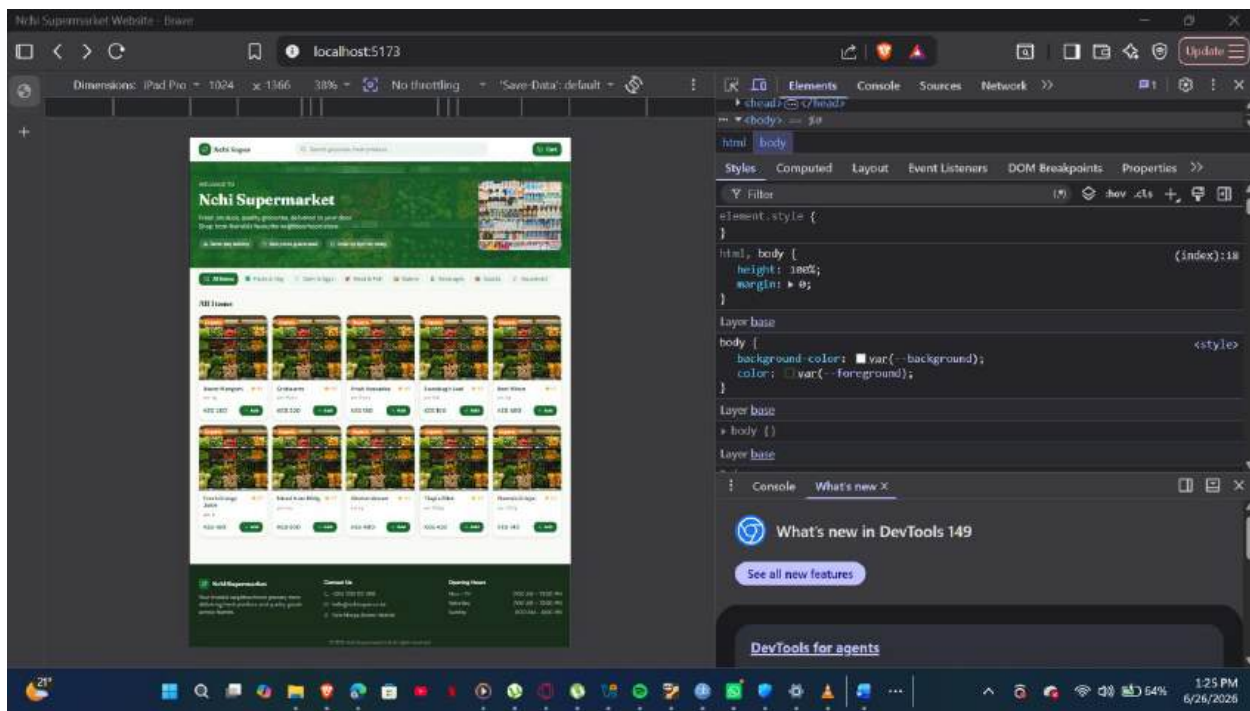


Fig 5 product_grid tablet.

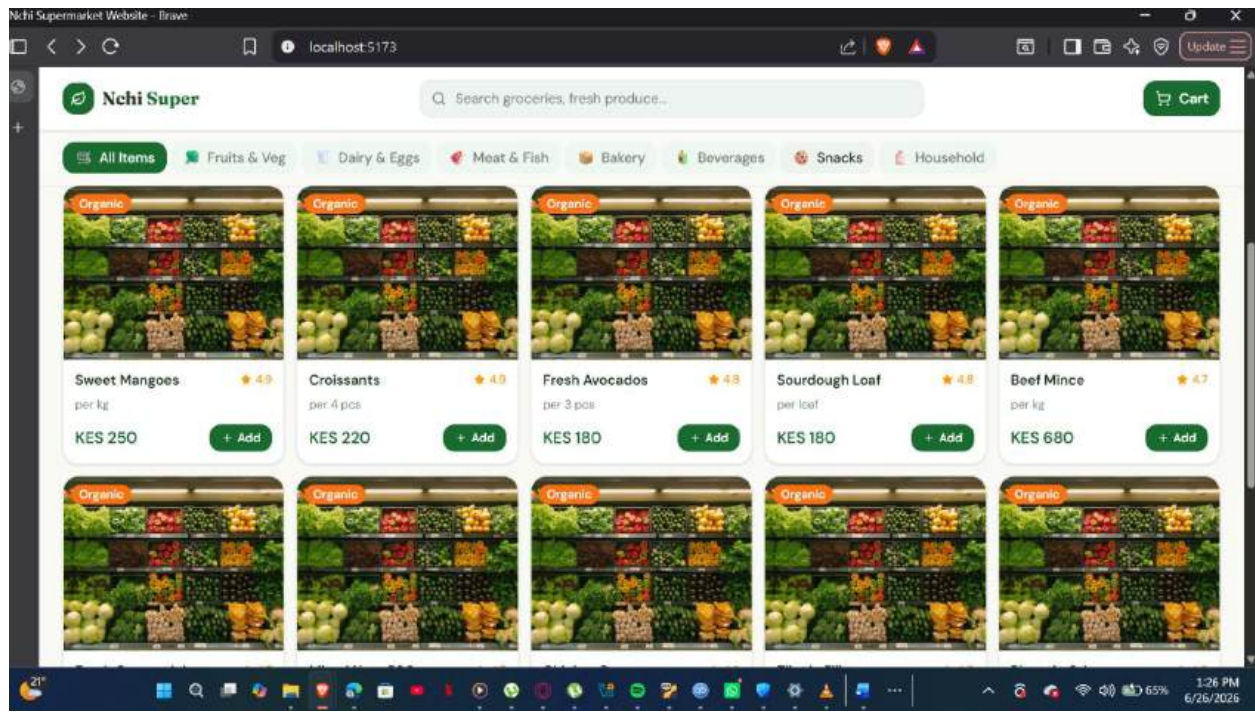


Fig 6 product_grid desktop

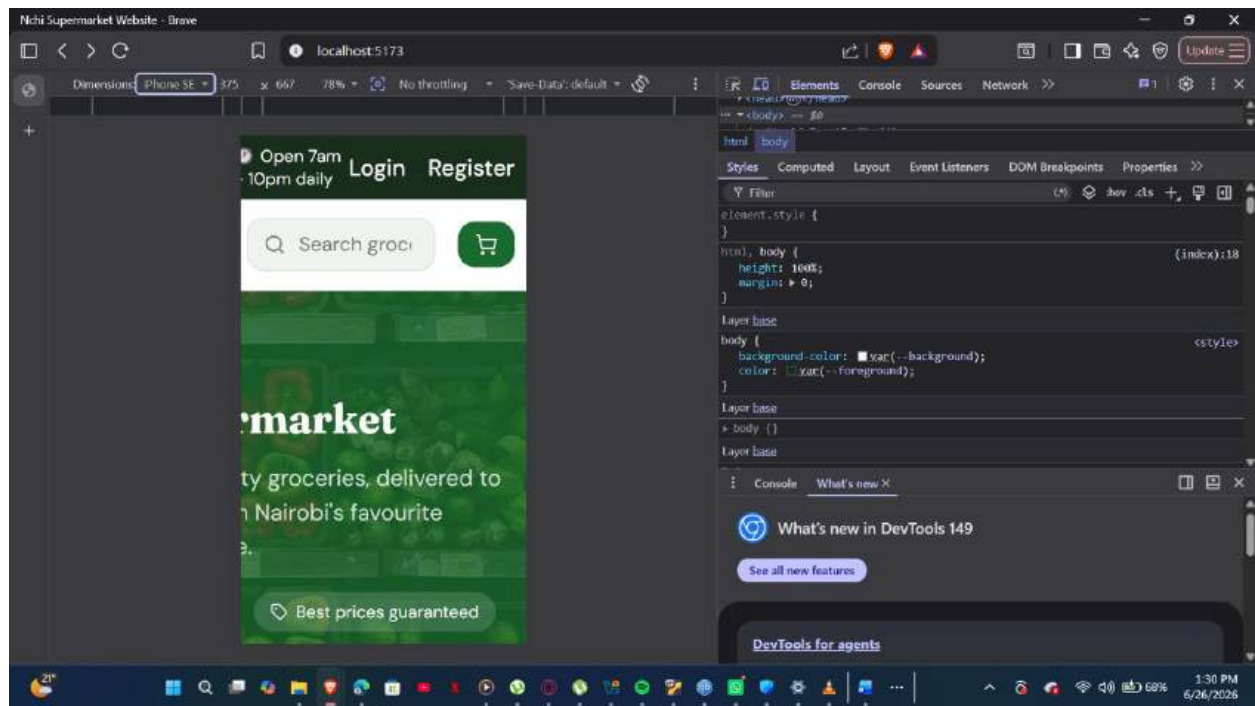


Fig 7 navigation mobile.

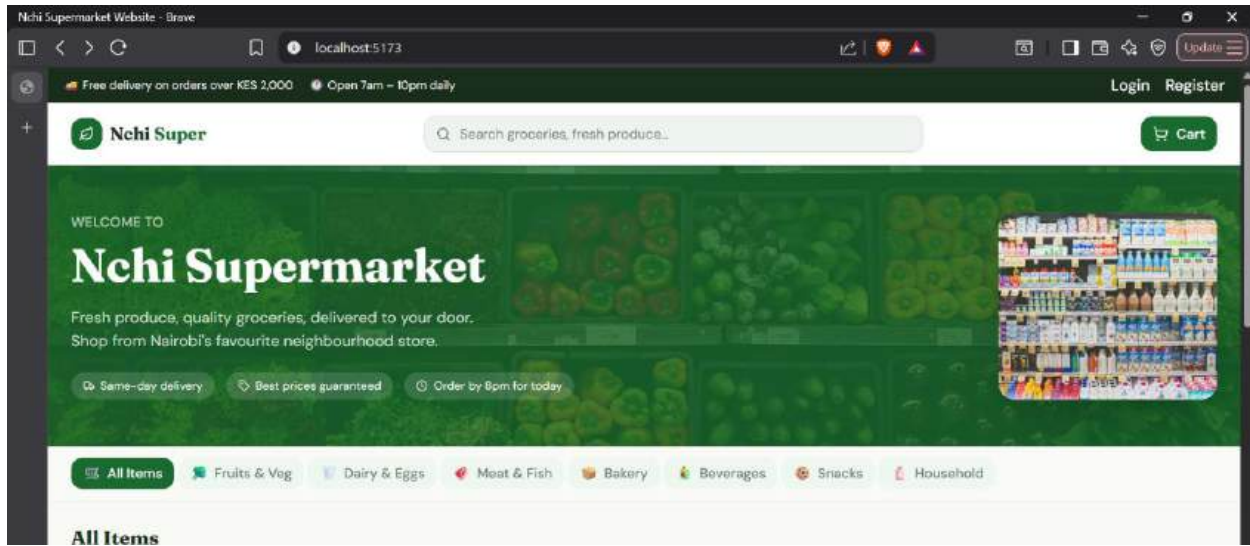


Fig 8navigation desktop

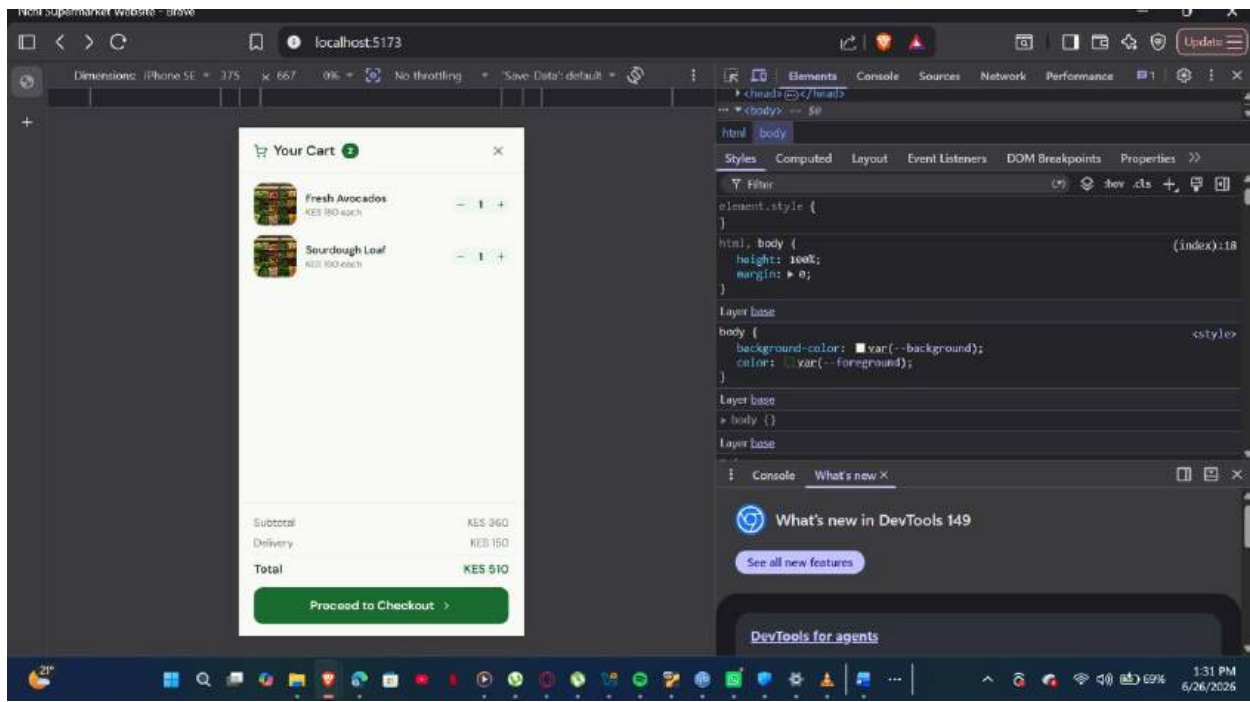


Fig 9cart mobile

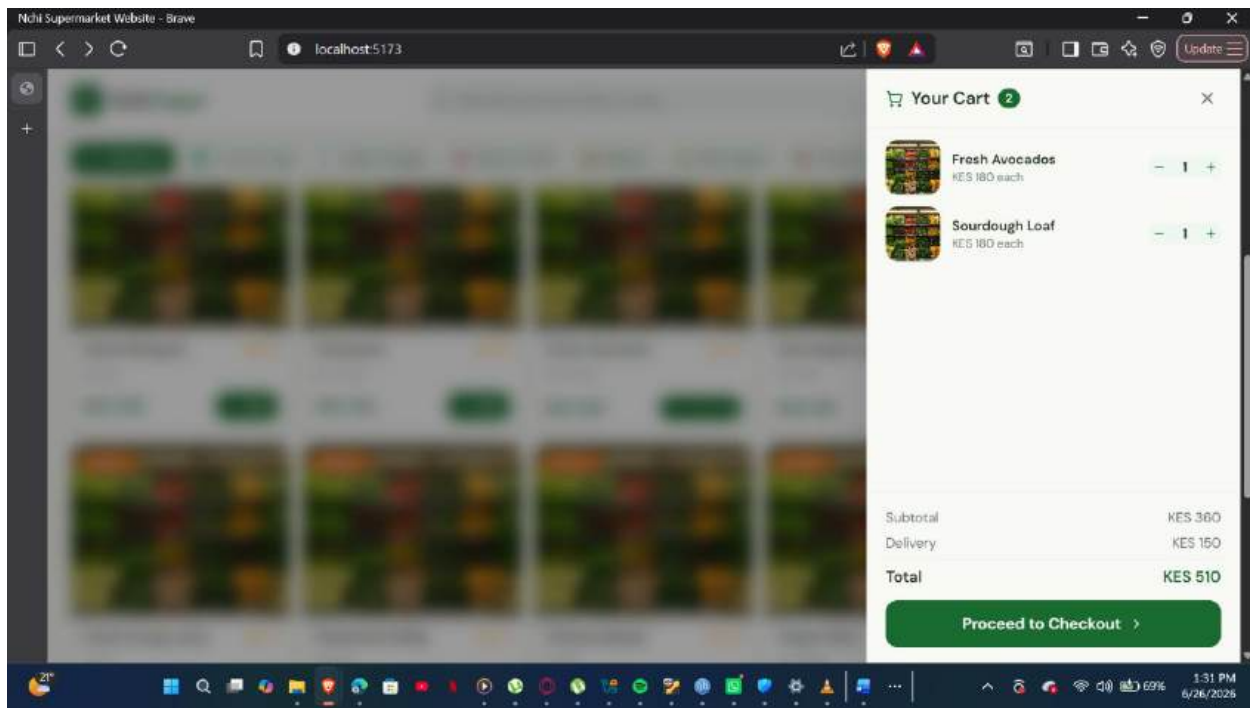


Fig 10cart desktop

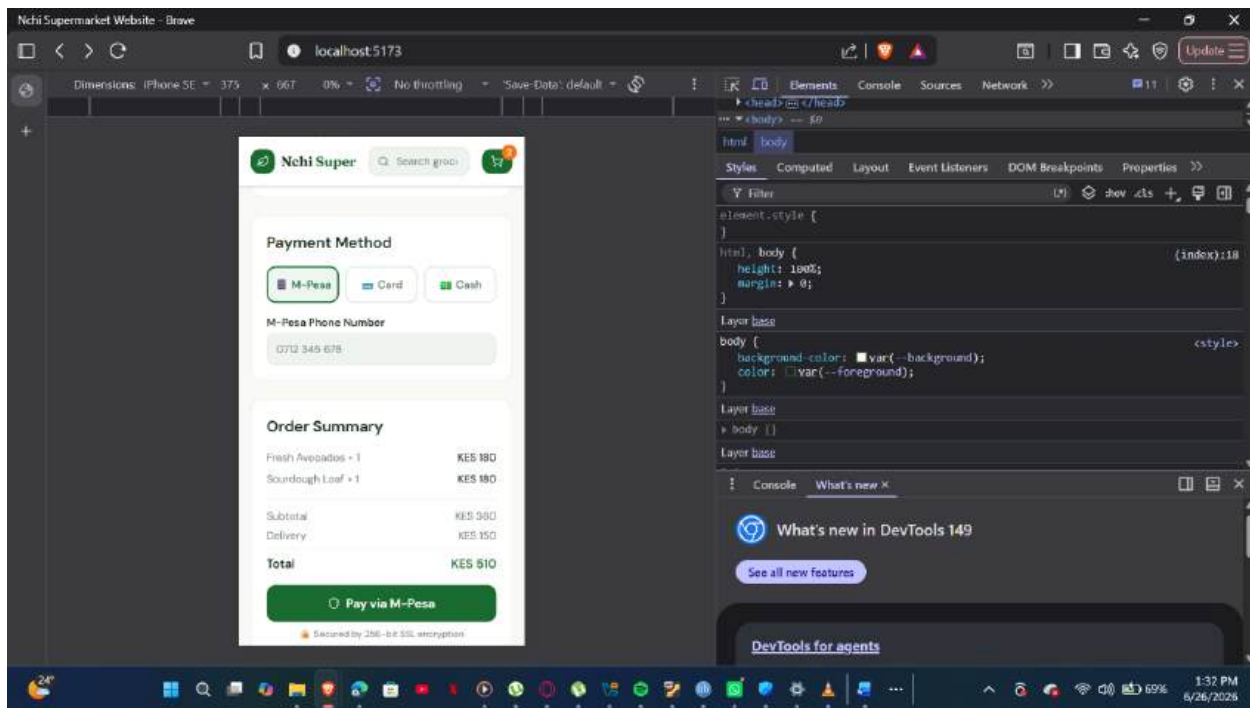


Fig 11checkout mobile.

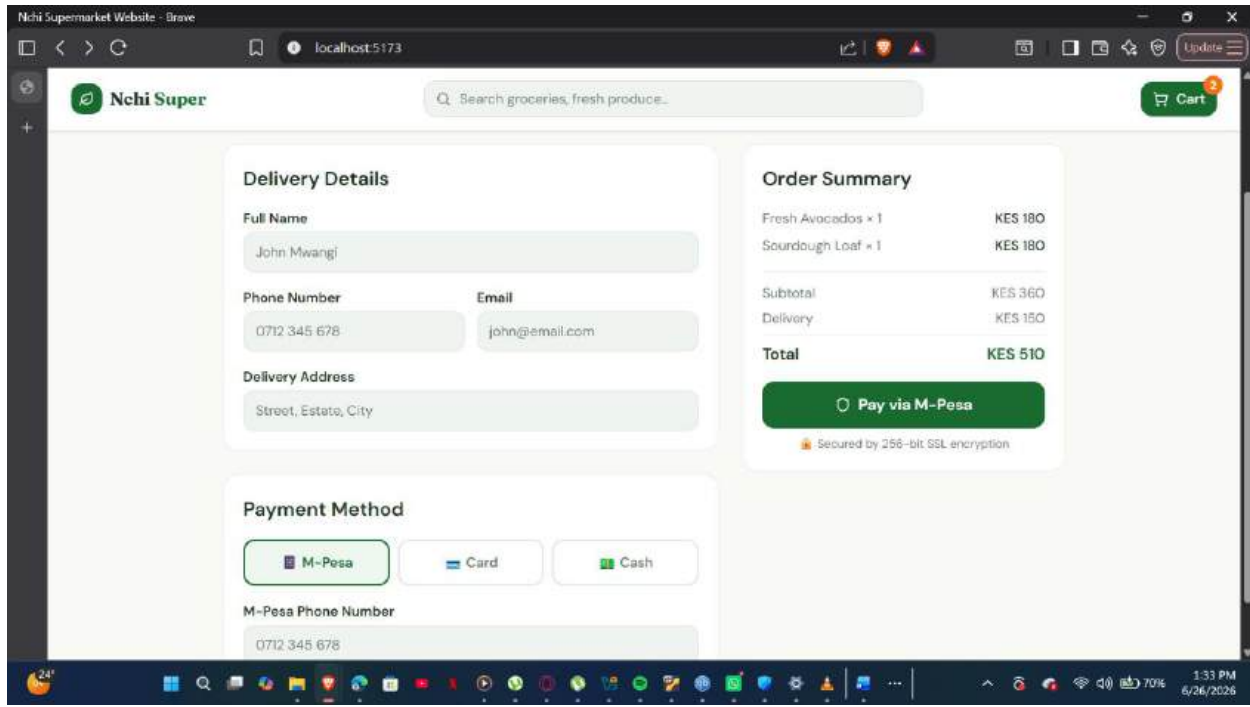


Fig 12desktop checkout

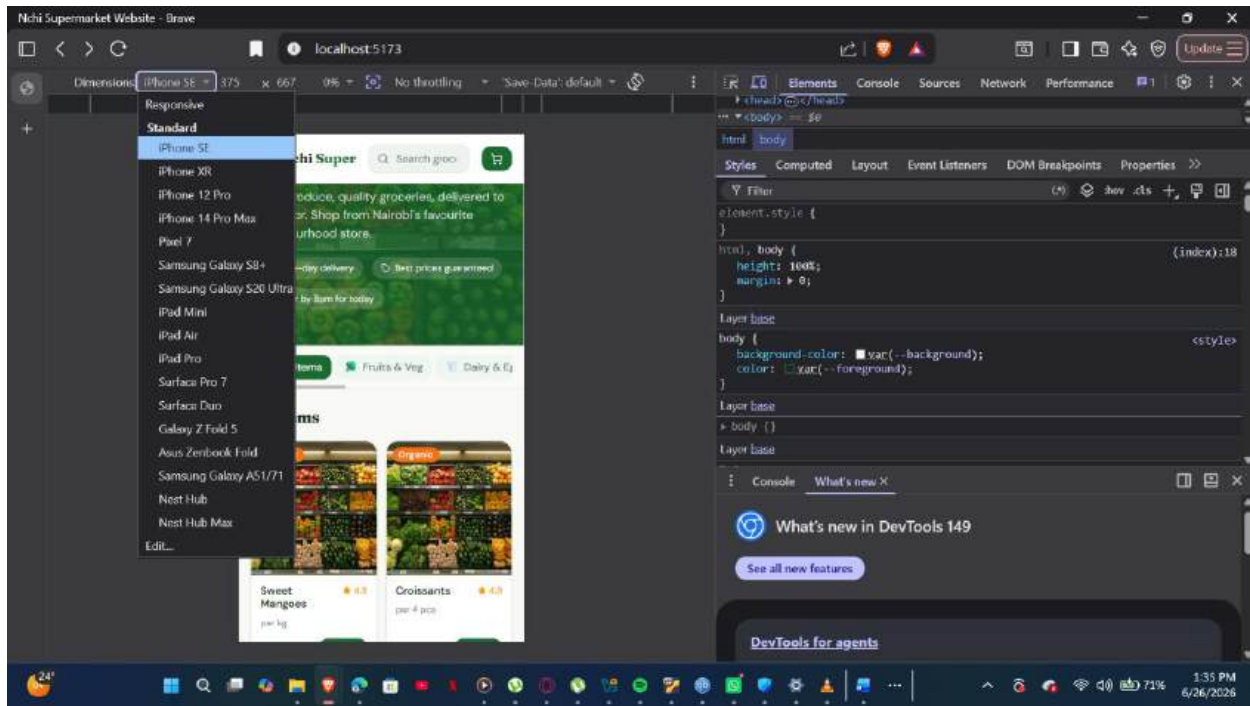


Fig 13responsive testing.

